

SLATEFALL: PAMC Operational Dossier

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Section 1. Identity, Background, and Public Status

1.1 Registry and Designations

The asset designated **SLATEFALL** is filed under the Pan-American Metahuman Cooperative (**PAMC**) Master Registry as **PAMC-A2-014**, registered 19 March 2019. Operational classification: **Class A-2 (Kinetic, Targeted)**. The codename was assigned by the PAMC Naming Subcommittee on 7 May 2019, derived from the asset's powers-emergence event on the eastern slate face of **Cerro Castillo**, Aysén Region, Chile.

Three additional designations remain in active rotation:

- **"Calva"** — informal in-cell handle used within Cell Halcón
- **"0-3-Castillo"** (zero-three-Castillo) — radio call sign for cross-cell coordination
- **"La Detención"** — adversary-side designation observed in three intercepted Concession-syndicate communications (2024); unofficial, not used internally

Her registry threat-to-handler rating is **0.7 / 5**, the lowest among PAMC Class A-2 operatives currently active.

1.2 Civilian Identity

The asset's civilian identity is **Capt. (ret.) Inez Yolanda Calvache-Renström**, born 8 September 1987 in Punta Arenas, Magallanes Region, Chile. Father: Coronel (ret.) Hernán Calvache-Mardones, Chilean Army military geologist (deceased 2017). Mother: Dr. Claire Renström-Verdugo, Franco-Chilean pediatrician (retired 2019; resides Viña del Mar).

The family relocated to Santiago in late 1999 following her father's posting to the Army's Centro de Estudios Geográficos. Calvache-Renström retained dual Chilean–French citizenship from birth; the French citizenship is currently administratively suspended pending PAMC liaison clearance, last reviewed Q3 2025.

1.3 Education

Period	Institution	Degree / Standing
2005–2009	Escuela Militar Bernardo O'Higgins, Santiago	Officer Cadet, class rank 14 / 162
2010–2012	Centro de Entrenamiento de Montaña (CEM), Río Blanco	Mountain Warfare Instructor Certification, Grade A
2012–2014	Universidad de Chile (part-time, military secondment)	MSc Mountain Engineering; thesis: glacier-ice anchor systems
2016	International Mountain Federation, Chamonix	Advanced Alpine Rescue Diploma

1.4 Pre-Powers Military Service

Calvache-Renström served in the Chilean Army for 14 years, from commissioning on 12 December 2009 through medical discharge on 28 February 2019. Rank at discharge: **Capitán** (Captain). Service distribution:

- 2009–2012: line officer, **Regimiento de Infantería Reforzado No. 23 "Carampangue"**, Coyhaique
- 2012–2014: platoon command, **Brigada de Operaciones Especiales "Lautaro"** (mountain-detachment cycle)
- 2014–2018: senior instructor, CEM Río Blanco — primary specialty in rope-rescue, ice-traverse, and high-altitude assault
- 2018: trial-deployment with the PAMC observer cell in Antofagasta — last operational posting before powers emergence

She participated in three named cross-border deployments under the **OEA-MAR (Organización de los Estados Americanos — Misión Andina de Rescate)** framework: Operación Vilcabamba (Peru, 2014), Operación Trapecio (Bolivia, 2016), Operación Cancha Carrera (Argentina, 2017).

1.5 Physical Description

- **Height:** 1.68 m
- **Weight:** 71 kg (operational); 69 kg standard
- **Build:** athletic, body-fat ~14%, pronounced upper-back musculature consistent with sustained pack-load training
- **Hair:** dark brown, cropped short (≤ 4 cm)
- **Eyes:** hazel-green
- **Skin tone:** olive
- **Distinguishing marks:**
 - 4.2 cm crescent scar over the left brow ridge (residual from Cerro Castillo, 2018)

- Partial nerve damage in the right thumb with surgical scar (Lima, 2020)
- Discreet tattoo on left ribcage: a small geodesic triangle (regimental keepsake)
- **Blood type:** A+
- **Dominant hand:** right (post-injury, cross-dominant for trigger use)

1.6 Languages

Language	Proficiency	Operational use
Spanish (Chilean)	Native	All operational
French	Fluent	Family correspondence, Quebec liaison
English	Fluent	PAMC official channels
Mapudungun	Working	Southern Chilean operations
Aymara	Conversational (acquired 2023)	Altiplano operations
Portuguese	Comprehension only	Brazilian liaison, occasional

1.7 Marital Status and Family

Married 4 November 2023 to **Dr. Catalina Renström-Verdugo**, pediatric oncologist at Clínica Las Condes, Santiago. (The spouses share a surname element by coincidence; no consanguineous relation.) No children. Spousal access to the asset's quarters is restricted to two named PAMC-cleared residences (Valparaíso and Santiago). Dr. Renström-Verdugo holds **PAMC Civilian Liaison Clearance Level 3**, granted 12 January 2024.

The asset's only surviving sibling is **Mauricio Calvache-Renström**, 34, civilian, vineyard manager in the Maipo Valley; uncleared, contact limited per Cooperative protocol.

1.8 Public Status

Unlike most PAMC Class A operatives, SLATEFALL is **publicly registered and publicly identified**. The following PAMC after-action reports have been declassified and released through the Cooperative's Lima press office:

Report ID	Date released	Incident
PAMC-PR-2021-006	14 July 2021	Antofagasta containment
PAMC-PR-2024-002	22 February 2024	Tarapacá compound raid
PAMC-PR-2025-011	9 December 2025	Potosí intervention

She has given two on-record interviews (both Spanish-language, broadcast Canal 13 Chile 2022 and TVN 2024). Public-facing media presence is otherwise restricted under PAMC **Protocol 4-Bravo**. She does not maintain personal social media accounts.

1.9 Psychological Profile Summary

Most recent full evaluation: 3 February 2026, conducted by Dr. Maite Borghi-Castelnuovo (PAMC Behavioral Health, Lima). Summary findings:

- High conscientiousness, low neuroticism
- Stress tolerance 92nd percentile relative to PAMC peer cohort
- Mild PTSD symptoms managed through structured-cognitive therapy since November 2020 (post-Antofagasta)
- No documented dissociative incidents
- Risk profile: low for unauthorized disclosure; low for unilateral action; moderate for cognitive fatigue under prolonged altitude operations
- Recommended supervisory cadence: monthly review, quarterly full evaluation

She is cleared for command authority over Cell Halcón and authorized for solo deployments not exceeding 14 calendar days.

Section 2. Powers, Abilities, and Documented Limits

2.1 Core Mechanic: Inertial Suspension

The asset's primary power, designated **Inertial Suspension (IS)** under PAMC scientific cataloguing, permits the targeted nullification of inertia in a single physical object within line-of-sight range. The effect was first observed during the powers-emergence event of 8 October 2018 (see §3.4) and has been continuously studied by the **Cooperative Bureau of Metahuman Physics (CBMP)** since March 2019.

The operative theory of mechanism — the **Calvache–Marrón Equations** — was developed jointly by the asset and CBMP physicist Dr. León Marrón-Saavedra and published in restricted form on 11 May 2022. The equations posit a transient local discontinuity in the conserved-momentum tensor, sustained by metabolic substrate drawn from the asset's central nervous system. The discontinuity is bounded in mass, duration, range, and recurrence, all of which have been experimentally calibrated.

When the effect is active, the suspended object:

- has effective inertial mass equal to approximately **0.4%** of its rest mass relative to the local Earth frame
- retains its gravitational mass (small objects still fall in earth gravity, but with no resistance to lateral force)
- does not interact mechanically with the asset herself (collision exemption)
- resumes its prior linear and angular momentum *exactly* upon termination (see **Tail Momentum**, §2.6)

2.2 Targeting Procedure

Activation requires a focused **line-of-sight acquisition** of the target, sustained for **1.2 seconds** baseline. This focus is mediated by visual attention; auditory or proprioceptive targeting is not possible. The asset describes the targeting period internally as "tuning the weight out."

Targeting failure rates by environmental condition:

Condition	Targeting failure rate
Dry, clear, daylight	0.4%
Light precipitation	2.1%
Heavy precipitation / snow whiteout	7.0%
Fog $\leq$ 12 m visibility	11.0%
Smoke (operational obscurant)	18.0%
Active strobing light source	26.0%
Total visual occlusion	100% (no activation possible)

Targeting is interrupted irreversibly by acoustic shock $\geq$ **135 dB** delivered within the 1.2 s window; the asset must restart acquisition from zero and incurs a 4.2-second neural recovery delay before re-attempting.

2.3 Mass Ceiling

The controlled mass ceiling under standard operational conditions is **240 kg**. This figure was established through CBMP testing across five quarterly cycles between 2019 and 2022 and has remained stable through 2026.

Recorded peak (uncontrolled): **380 kg**, observed during the 2018 trigger event when she suspended a fragmenting block of slate during her fall. The trigger event remains the sole recorded instance above 250 kg and is not reproducible under controlled conditions.

Above the 240 kg ceiling, attempted activation produces one of three outcomes:

1. Activation failure with neural feedback (mild migraine onset) — 41% of cases
2. Partial suspension of a fraction of the target (e.g., one face of a larger object) — 36% of cases
3. Full activation followed by **Echo Lock** failure mode (see §2.13) — 23% of cases

2.4 Duration

Baseline maximum sustained duration of a single suspension: **0.6 seconds**. This figure is hard-capped under normal physiological conditions and is independent of target mass within the 240 kg ceiling.

Duration extension is achievable via pharmacological augmentation. Under therapeutic dose of **Calexin-7** (240 mg, twice daily, see §4.9), tested duration extends to **1.1 seconds**. The Cooperative does not authorize Calexin-7 doses beyond therapeutic levels; off-label dosing of 360 mg in 2022 yielded 1.4 s duration but produced severe vasoconstrictive side-effects and is prohibited under Cooperative Medical Standing Order 11.

2.5 Range

Effective operational range: **22 meters** measured from the asset's nasion to the target's center of mass. Theoretical maximum: **41 meters**, with substantially degraded reliability; targeting resolution declines past **28 meters** at a rate of approximately 1.4% failure increase per additional meter.

Range is not affected by intervening *transparent* obstacles (glass, ice, polymer film); it is interrupted by any opaque obstacle within the visual cone regardless of material thickness. The effect propagates instantaneously at all measured ranges (sub-microsecond resolution).

2.6 Tail Momentum

When the suspension ends, the target resumes its prior linear and angular momentum **vector** exactly — direction, magnitude, and rotation. This property is termed **Tail Momentum** and underpins the **Tail-Strike** combat technique (see §5.4).

Tail Momentum has been measured to within $\pm 0.03\%$ accuracy in CBMP impulse-recovery trials. The only documented deviation occurred during the **Antofagasta incident** (May 2021), when an asymmetric thermal gradient applied to a suspended target produced a 2.8% lateral drift; this anomaly has not been replicated under controlled conditions.

2.7 Cooldown Matrix

The recurrence-rate limits on Inertial Suspension are governed by three interleaved cooldowns:

Cooldown type	Duration	Trigger
Per-target cooldown	11.0 s	Same physical object cannot be re-suspended within this window
Global activation cooldown	2.4 s	Any two activations on any targets must be separated by at least this interval
Stamina-window cap	Max 3 activations per 60 s	Underlies the "Three-Two-One Rule" (§5.2)

Violations of the per-target cooldown produce **null activation** (no effect, full metabolic cost). Violations of the global cooldown produce **half-duration** suspension (0.3 s baseline) and elevated migraine onset risk.

2.8 Contact Exclusion

The asset cannot suspend an object that she is in direct physical contact with at the moment of trigger. The contact exclusion applies to:

- Held objects (e.g., she cannot suspend a baton she is currently gripping)
- Worn equipment
- Surfaces she is standing on or pressed against
- Other personnel she is touching

She *can*, however, suspend an object that she releases within **~0.3 seconds** prior to trigger. The standard tactical drill known as **"Drop-Tune"** exploits this: she releases a throwing baton, lets it travel briefly, then suspends it mid-flight for repositioning (see §5.5).

2.9 Self-Suspension

The asset can target her own body for Inertial Suspension, provided the contact-exclusion rule is satisfied (i.e., she is not currently in contact with another large mass at the trigger instant). Mid-air self-suspension is a documented life-saving application — the 2018 trigger event was itself a mid-fall self-suspension that arrested a 47-meter descent for the 0.6-second window required to reorient.

Self-suspension during ground contact requires brief in-air transitions and is rarely tactically applicable. Self-suspension while gripping a weapon is forbidden under standing orders due to the contact rule (see §5, Directive 7).

2.10 Secondary Ability: Drift Read

Drift Read is a passive, involuntary precognitive sense awakened during the Antofagasta incident, 17 May 2021. It manifests as a tactile pre-perception of incoming projectile vectors within approximately **3 meters** of the asset's body, perceived as a "draft" or directional pressure sense on the skin.

Drift Read parameters:

Parameter	Value
Range	3.0 m, hard-capped
Lead-time	~0.30 s before projectile arrival
Applicable projectiles	Any solid object travelling ≥ 8 m/s
Inapplicable to	Liquids, plasma, electromagnetic radiation, sound pressure
Active during sleep	Yes, degraded by ~60%
Voluntary disengagement	Not possible

Drift Read does not directly enable evasion; it provides directional information that the asset processes consciously and exploits via reflex or by triggering a fast Inertial Suspension on the incoming object.

2.11 Tertiary Ability: Hold

Hold is a low-mass, extended-duration variant of Inertial Suspension available for objects up to **2 kg**. It permits sustained suspension for up to **4.0 seconds** at a metabolic cost approximately equal to a single full-duration standard suspension.

Hold is restricted to objects she has already successfully suspended once within the past **90 seconds**; Hold is a "re-tune" of an existing inertial signature, not a primary engagement. It cannot be applied to living tissue and cannot be applied to self.

Operational use: keeping a thrown baton mid-air to choreograph a multi-baton kinetic field (the **Suspended Field** technique, §5.6).

2.12 Prohibited Targets

The following are categorically excluded from Inertial Suspension under all known conditions:

Prohibited target	Reason
Liquids in flow >4 L/s	Targeting cannot resolve a coherent center of mass
Plasma	No bound inertial signature
Rotating bodies > 1,800 rpm	Angular-momentum binding fails
Biological tissue exceeding 12 kg	Cooperative ethical prohibition <i>and</i> inconsistent results
Active explosive ordnance post-ignition	Energy release exceeds binding
Quantum-correlated metahuman effects	Unstable interaction; prohibition added Q2 2024
Objects within 0.4 m of the asset's skin	Contact-exclusion field

2.13 Echo Lock Failure Mode

Echo Lock is a neurological seizure-equivalent observed twice in operational record:

- 19 November 2020 (training environment, Lima)
- 14 August 2023 (combat environment, Salta)

Symptoms: a 4-second targeting freeze in which the asset cannot acquire new targets, cannot voluntarily release a current suspension, and experiences pronounced cervical spasm. Recovery is autonomous; no permanent neurological damage has been documented. Triggering conditions include:

- Attempted suspension above the 240 kg ceiling
- Acoustic shock during the targeting window (≥ 145 dB)
- Severe hypoglycemia (blood glucose < 58 mg/dL) during sustained operation

Echo Lock occurrence remains rare (**2 events** / approximately **3,200 logged activations** through 31 December 2025).

2.14 Metabolic Cost

Operational metabolic demand at peak operational rate (3 activations/min sustained):

- Caloric burn: **~2,100 kcal per operational hour**
- Sodium loss: **1.4 g/hr**
- Glycogen depletion: complete within **38 minutes**
- Required carbohydrate intake during sustained operation: **80 g/hr** complex, plus emergency **25 g** fast-acting
- Core-temperature elevation: +0.6 °C per operational hour at peak rate

Field nutrition protocol mandates hypertonic glucose gel every 30 minutes during active engagement and a standardized electrolyte solution (PAMC-issued formulation **Bofedal-3**, 740 mL/hr) between activations.

2.15 Neurological Tax

Chronic neurological consequences observed:

- Cervical migraines, frequency 3–5 per week without prophylaxis
- Pharmacological management: **Calexin-7** 240 mg twice daily (see §4.9)
- Sleep latency increase of ~14 minutes baseline
- REM-cycle disruption on operational nights (~22% reduction)
- Cumulative biological aging correlate: **telomere shortening +2.4 years per operational year** (most recent measurement Q4 2025)
- No documented cognitive decline; cognitive performance assessment Q4 2025 within prior baseline

2.16 Documented Vulnerabilities

In addition to the Echo Lock trigger conditions, the following operational vulnerabilities are formally documented:

- **Acoustic shock ≥ 135 dB** during targeting interrupts acquisition (§2.2)
- **Sustained sensory deprivation** (visual occlusion > 40 minutes) elicits transient power suppression
- **Hypothermia core < 34.6 °C** disables Inertial Suspension entirely
- **Hyperventilation** (rapid > 34 breaths/min for > 90 s) degrades targeting by 18%
- **EM-pulse exposure above 0.4 T** does not directly disable the power but causes secondary nerve dysfunction
- **Concurrent metahuman countermeasures** documented twice (see §7.3 — Tropopausa; §7.10 — Mar Inverso)

2.17 Power Performance Statistics (2019–2025)

Cumulative operational data drawn from PAMC field logs and CBMP debrief telemetry:

Metric	Value
Total logged activations	3,217
Successful activations	3,074 (95.6%)
Targeting failures	102 (3.2%)
Mid-activation aborts	41 (1.3%)
Average target mass per activation	38.2 kg
Median activation duration utilized	0.42 s
Mid-air self-suspensions logged	47
Tail-Strike applications logged	312
Suspended Field deployments	89
Echo Lock incidents	2
Operational hours under power	1,148

2.18 Comparative Classification

Within the PAMC roster, SLATEFALL occupies a narrow operational niche. Of 217 currently registered Cooperative operatives:

- 38 are Class A (Kinetic), of whom 12 are Class A-2 (Targeted)
- Only 4 Class A-2 operatives possess an inertial- or momentum-related primary power
- SLATEFALL is the only documented case of true inertial *nullification* (the others act on energy transfer rather than mass)
- Comparable adversary capability is suspected in only one open file: **Mar Inverso** (§7.10), projected Tier 5

2.19 Research and Re-Evaluation Schedule

Annual re-evaluation of power parameters is mandated under PAMC Scientific Standing Order 4. The next scheduled CBMP full assessment is 11–14 June 2026 in Lima. Interim quarterly checks are performed by Dr. Solène Bauer-Cortez at Cuartel Valparaíso.

2.20 Documented Anomalies and Unresolved Mechanics

Four anomalies remain formally classified as **Unresolved Mechanics** by CBMP and are flagged for future research cycles:

1. **The Antofagasta Drift (May 2021):** the singular 2.8% lateral deviation in Tail Momentum recorded during the Drift Read awakening event. Hypothesis: asymmetric thermal gradient from a nearby refinery flare. Replication attempts in CBMP labs (4 attempts, 2021–2024) all failed to reproduce the effect.
2. **The "Twin-Lock" hypothesis:** during the 14 August 2023 Salta engagement, telemetry recorded a 0.21-second period in which two simultaneously suspended objects exhibited *interlocked* termination — both released within 14 milliseconds of each other despite separate activation timestamps. No mechanism has been proposed.
3. **The 2024 Coquimbo dampening:** during the 4th Tropopausa engagement (Coquimbo, May 2024), the asset's targeting failure rate dropped to **0.0%** across 7 activations — well below the 0.4% baseline. Hypothesis: synergistic interaction with Tropopausa's pressure field. Unverified.
4. **REM-cycle bleed:** during REM sleep, the asset's Reset Bracers occasionally register low-amplitude targeting-precursor signals despite no conscious activation. Frequency: ~3 per night. No external effects have been observed; the phenomenon may be a subconscious artifact rather than a true power expression.

These anomalies are reviewed each annual cycle and are not currently considered operational risks.

2.21 Comparative Counter-Threat Profile

Should the asset face an inertial-manipulation adversary (e.g., the projected Mar Inverso, §7.10), CBMP has modeled the engagement profile under three scenarios:

Scenario	Adversary capability	Projected outcome
Mass <i>multiplication</i> (inverse symmetry)	×50 inertial increase, 240 kg ceiling, 0.6 s duration	Tactical stalemate; mutual cooldown exhaustion likely

Scenario	Adversary capability	Projected outcome
Mass <i>imprint</i> (locking momentum onto target)	Inert object pinned in place	SLATEFALL operationally suppressed if pinned
Mass <i>transfer</i> (redistributing inertia between objects)	Coupled-object momentum manipulation	Unknown; high risk of cascade failure

The third scenario is the most concerning and is the rationale behind the Director-General-only authorization requirement for Mar Inverso engagement.

2.22 Power-Onset Pre-Activation Signatures

CBMP has documented three measurable pre-activation physiological signatures detectable approximately **0.4 seconds before** the asset initiates targeting:

- Pupillary dilation increase of 11–14% (target-side pupil only)
- Skin conductance spike of 2.6 μ S on left thenar eminence
- Spinal-cord vibrational micro-signature, 78 Hz, recorded via Reset Bracer accelerometers

These signatures are not detectable by external observers without specialized instrumentation, but the Cooperative classifies them as **Operational Tell-Tale Indicators (OTTI)** and rotates the asset's pre-engagement breathing protocol quarterly to mitigate predictive exploitation by adversaries.

Section 3. Origin and Key Historical Events

3.1 Early Life (1987–2005)

Inez Yolanda Calvache-Renström was born on 8 September 1987 at the **Hospital Naval de Punta Arenas**, in conditions noted as "uneventful" on her birth record. Her early childhood, until age 12, was spent in the Magallanes Region, where her father served as a regional army geologist and her mother operated a small pediatric practice. The family resided at Calle Errázuriz 1422, Punta Arenas — a residence later sold in 2001.

In late 1999, following her father's reassignment to the **Centro de Estudios Geográficos** in Santiago, the family relocated to the capital. Calvache-Renström completed secondary education at the **Liceo Manuel de Salas** between 2001 and 2004, graduating with academic honors in geography, mathematics, and physical education. She represented the school in two consecutive national alpine-skiing championships (2003 and 2004), placing 7th and 4th respectively in the U18 women's giant slalom.

A 2003 field trip to **Glaciar Exploradores** in the Aysén Region is recorded in subsequent autobiographical interviews as a formative experience.

3.2 Military Cadet and Junior Officer Years (2005–2014)

She entered the **Escuela Militar Bernardo O'Higgins** on 4 March 2005. Cadet rank progression and notable cycle assignments are summarized in §1.3 and §1.4. She graduated 12 December 2009 with class rank **14 of 162**, receiving the **Espada de Honor de Geografía Militar** for top scores in topographic surveying.

Junior officer service distribution (2009–2014):

Period	Posting	Rank
Jan 2010 – Apr 2012	Regimiento "Carampangue," Coyhaique	Subteniente
May 2012 – Jul 2012	CEM, Río Blanco (instructor pipeline)	Subteniente
Aug 2012 – Sep 2013	Brigada "Lautaro," mountain detachment	Teniente
Oct 2013 – Dec 2014	Brigada "Lautaro," senior platoon role	Teniente

On 19 March 2014 she was selected for instructor secondment at **CEM Río Blanco**, a posting traditionally indicating fast-track promotion intent.

3.3 Senior Instructor Period (2014–2018)

From 2014 through mid-2018, Calvache-Renström served as a senior instructor at the Centro de Entrenamiento de Montaña, attaining the rank of **Capitán** on 1 March 2017. During this period she led 11 instructor cycles, certified 142 mountain warfare graduates, and authored two internal Army doctrine annexes:

- Annex M-14: *Rope-Rescue Procedures Above 4,500 m*
- Annex M-16: *Ice-Anchor Selection in Mixed-Substrate Faces*

She received the **Cruz al Mérito Militar (Tercera Clase)** on 7 September 2017 for the successful extraction of a stranded civilian climbing party from **Volcán Tupungato** during inclement weather.

3.4 The Cerro Castillo Incident (8 October 2018)

The asset's powers-emergence event occurred on **8 October 2018** at approximately **14:42 local time** on the eastern face of **Cerro Castillo** (47°15'S, 72°06'W), elevation ~2,318 m at incident point. She was conducting a solo reconnaissance ascent in preparation for a CEM instructor cycle, climbing a Grade IV-5 mixed slate-and-ice route.

Sequence of events as reconstructed from her helmet recorder, recovered ice tools, and post-incident geological survey:

1. **14:40:11** — Anchor screw #4 was placed at 2,318 m elevation.
2. **14:41:37** — Rope-end ice tool placement failed in unconsolidated slate.
3. **14:41:52** — A 1.2 m × 0.8 m slate block detached from the face, striking her left brow ridge.
4. **14:41:54** — Anchor screw #4 sheared; she entered free-fall.
5. **14:41:58** — During the fall, she experienced an involuntary first activation of Inertial Suspension, arresting the descent of a fragmented 380 kg slate mass adjacent to her body for an estimated 0.6 s, allowing reorientation.
6. **14:42:00** — Self-suspension event arrested her own 47-meter fall for the same 0.6-second window; she resumed motion at reduced trajectory and impacted a snowfield rather than rock.
7. **14:42:14** — Loss of consciousness, traumatic.

She was located by a CEM rescue patrol 4 hours 21 minutes later, evacuated by helicopter to **Hospital Regional de Coyhaique**, and transferred 11 October 2018 to **Hospital Militar Santiago**. Injuries: closed-

head trauma, right thumb fracture with nerve damage, mild hypothermia, multiple contusions. No fatalities; no other personnel were on the mountain at the time.

3.5 Recovery and Powers Confirmation (October 2018 – February 2019)

During her hospitalization, anomalous events were observed by attending nursing staff between 24 October and 11 November 2018: small objects in the immediate vicinity of her bed exhibited transient "weightless" motion correlating with REM cycles. On 14 November, a controlled observation confirmed reproducible inertial suspension of a 200-gram steel object at her request.

She was transferred to the **Lima Metahuman Assessment Facility (LMAF)** on 28 January 2019. PAMC initial assessment classified her capability as a novel Class A-2 phenomenon. Medical discharge from the Chilean Army was finalized **28 February 2019**.

3.6 PAMC Recruitment and Probationary Period (March 2019 – November 2020)

She signed the Cooperative Service Agreement on **12 March 2019**. Codename SLATEFALL assigned 7 May 2019. Probationary deployment cycle:

- **19 March 2019 – 22 August 2019**: residential training, Lima HQ
- **23 August 2019 – 14 January 2020**: paired-cell rotations under senior operative **MIRADOR** (Cpt. (ret.) Estefania Pizarroso-Aldama)
- **15 January 2020 – 19 November 2020**: independent probationary deployments, Atacama Sector

The **19 November 2020 training incident** at Lima HQ produced the first Echo Lock event (see §2.13). Investigation concluded the cause was an unauthorized peer-test attempt to suspend a 320 kg test sled. Subsequent training discipline tightened; the mass ceiling protocol was formalized.

3.7 The Antofagasta Incident and Drift Read Awakening (17 May 2021)

On **17 May 2021** during a containment operation in the port district of Antofagasta (PAMC Case 2020-0048, see §9.2), the asset suffered a small-arms ambush from elevated cover. During the engagement, she experienced an involuntary precognitive perception of incoming projectile vectors — the first manifestation of what would be designated **Drift Read** (§2.10). The ability has remained continuously active since.

The Antofagasta operation produced a Successful classification but resulted in a 14-day medical leave for the asset and the documented onset of post-traumatic symptoms (§1.9).

3.8 Tropopausa First Engagement and the Mendoza Pyrrhic (June 2022)

On **3–5 June 2022**, the asset engaged the metahuman adversary **Tropopausa / Eliseo Mardones** for the first time, in the **Mendoza Free Trade Zone** (Case 2022-0123, §9.4). The engagement is classified **Pyrrhic**: Tropopausa was contained, but PAMC suffered 3 civilian and 1 operative casualty (Cpl. Adolfo Yáñez-Ríos, civilian medic, Cell Cóndor).

The Mendoza engagement produced revisions to PAMC engagement protocols against atmospheric-pressure manipulators and prompted the formal articulation of the **Doctrine of Sequential Suspension** (§5.1) on 19 June 2022.

3.9 Cell Halcón Formation and Marriage (2022–2023)

Following Mendoza, the asset was promoted to **Field Officer, Grade 2** on **1 August 2022**, with command authority over a newly-formed 4-operative cell designated **Cell Halcón**. Cell composition was finalized 11 October 2022 (see §6.2).

She married Dr. Catalina Renström-Verdugo on **4 November 2023** in a private civil ceremony in Viña del Mar. Cooperative liaison clearance for the spouse was processed in January 2024.

3.10 Salta Echo Lock and Quebrantadero (August 2023)

On **14 August 2023**, during Case 2024-0202 in Salta Province, Argentina (see §9.7), the asset experienced her second documented **Echo Lock** event while attempting suspension of a 280 kg vehicular component. The Salta engagement against **Quebrantadero / Wilma Sandoval-Iqui** produced significant civilian casualties (11 confirmed) and is classified **Catastrophic**. A formal Cooperative inquiry concluded in March 2024 with no censure of the asset but with three recommended protocol changes, all subsequently adopted.

3.11 Lead Officer Status and 2024–2025 Operations

On **9 February 2024**, the asset was promoted to **Lead Officer (Field, Grade 3)** with expanded command authority. During 2024 and 2025 she led 23 logged operations, including:

- Tarapacá Concession compound raid (Case 2024-0034)
- Tallinn-Asunción intelligence rotation (joint with European liaison)
- Lima Audiencia counter-audit (Case 2025-0019)
- Potosí Cabrestante neutralization (Case 2025-0156)

Cumulative operational hours under power during 2024–2025: approximately 612.

3.12 Current Deployment (2026)

As of revision date (11 April 2026), the asset is engaged in **Case 2026-0008**, classification restricted, with operational territory spanning the Bolivian altiplano and northern Chilean copper belt. The case is open and ongoing; specifics are withheld from this revision per PAMC Compartmentalization Order 3-Charlie.

3.13 Master Timeline Reference

Date	Event
8 Sept 1987	Born, Punta Arenas
12 Dec 2009	Commissioned, Chilean Army
1 Mar 2017	Promoted to Capitán
8 Oct 2018	Cerro Castillo trigger event
28 Feb 2019	Medical discharge from Chilean Army
12 Mar 2019	PAMC service agreement signed

Date	Event
7 May 2019	Codename SLATEFALL assigned
19 Nov 2020	First Echo Lock event (training)
17 May 2021	Drift Read awakens, Antofagasta
11 May 2022	Calvache–Marrón Equations published (restricted)
3–5 Jun 2022	Tropopausa first engagement, Mendoza
19 Jun 2022	Doctrine of Sequential Suspension articulated
1 Aug 2022	Promoted to Field Officer, Grade 2
11 Oct 2022	Cell Halcón finalized
14 Aug 2023	Second Echo Lock, Salta
4 Nov 2023	Marriage to Dr. Renström-Verdugo
9 Feb 2024	Promoted to Lead Officer (Field, Grade 3)
11 Apr 2026	Current dossier revision date

Section 4. Equipment, Gear, and Specialized Technology

4.1 Overview: PAMC Class A-2 Standard Field Kit

All PAMC Class A-2 operatives draw from a standardized issue under **Cooperative Logistics Standing Order 7**, customized per operative profile. SLATEFALL's customized loadout — designated **Loadout Profile HALC-014** — is calibrated for high-altitude, kinetic-targeted engagements and weighs **18.4 kg** at full operational configuration, excluding consumables.

Loadout profiles are recertified annually by the **PAMC Logistics Division (Valparaíso)** and the operative's commanding handler (currently **Director Salvatierra**, see §6.4). The most recent certification of HALC-014 occurred on 22 March 2026.

4.2 Field Uniform: Andina-7 Hardshell

Primary uniform layer is the **Andina-7 hardshell**, manufactured by **Tejidos Cordillera S.A.** (Talca, Chile) under PAMC contract. Specifications:

Parameter	Value
Total weight	3.8 kg (size M)
Outer shell	Tri-laminate aramid-nylon, 240 g/m ²
Insulating layer	Detachable, 110 g/m ²
Operating temperature	−40 °C to +45 °C
Pockets	14 (including 2 hidden)

Parameter	Value
Color profile	PAMC Andean Pattern M-2 (mixed grey/ochre, IR-suppressive)
Service life	380 operational days
Replacement cost (PAMC procurement)	USD \$4,200

The Andina-7 incorporates an integrated **biometric strain panel** at the cervical spine that interfaces with the Reset Bracers (§4.6) to provide real-time fatigue telemetry. The shell is rated to IIIA ballistic standard at chest and back panels; the integrated hood is rated to IIA.

4.3 Footwear: Cordillera Mk-IV

The **Cordillera Mk-IV** combat-mountain boot is the standard PAMC issue for Andean operations:

Parameter	Value
Weight per boot	980 g (size 38)
Manufacturer	Calzados Pirque, Santiago
Sole	Vibram-compound, 5.2 mm lug depth
Crampon compatibility	Universal (semi-automatic and step-in)
Insulation	200 g Thinsulate equivalent
Sole MTBF	320 km
Lacing	Speed-lock with redundant secondary tie

The Mk-IV variant introduced in 2023 includes a discreet **sole-cavity micro-cache** (volume 14 mL) used for emergency stimulant tablets.

4.4 Primary Kinetic Implement: Tungsten-Core Throwing Batons

The asset's signature kinetic implement is a set of six **tungsten-core throwing batons**, machined by **Talleres Magallanes** to PAMC specification **TC-340/B**:

Parameter	Value
Number carried (standard)	6
Mass per baton	340 g
Length	22.0 cm
Diameter	18 mm
Core	Tungsten-alloy, 96% W
Outer shell	Cold-blued steel, knurled grip

Parameter	Value
Balance point	Geometric center ± 0.4 mm
Engraved identifier	"HALC-014 / B-1" through "B-6"
Replacement cost per unit	USD \$940

The batons are designed for use *in conjunction with* Inertial Suspension. The standard tactical sequence — throw, mid-flight suspension, repositioning, release — is the basis of the **Drop-Tune** (§5.5) and **Tail-Strike** (§5.4) techniques. The asset reports a personal throwing accuracy of 88% within 12 m and 64% within 22 m without suspension assistance.

Loss record: of an original 24 batons issued since 2019, 18 remain in active service; 4 were lost in operations (Antofagasta ×1, Mendoza ×2, Tarapacá ×1) and 2 were retired due to material fatigue.

4.5 Sidearm: Bersa-Nogales TX-9

Issued sidearm is the **Bersa-Nogales TX-9** compact pistol, a PAMC-spec firearm produced in Argentina under joint Bersa–Nogales licensing. Specifications:

Parameter	Value
Caliber	9×19 mm
Magazine capacity	13 rounds
Weight, loaded	720 g
Weight, unloaded	580 g
Length	175 mm
Barrel length	91 mm
Trigger pull	2.4 kg, single-action after first round
Effective range	50 m

The asset's TX-9 (serial NGS-2229-AC) is the secondary engagement weapon; she fires it on average 14 rounds per operational year and prefers kinetic-baton engagement when range permits.

4.6 Reset Bracers

The **Reset Bracers** are paired wrist-mounted neural-telemetry units providing real-time targeting-fatigue data via a proprietary **galvanic-skin / heart-rate variability** algorithm. Engineered jointly by CBMP and Lt. Solène Bauer-Cortez:

Parameter	Value
Weight per bracer	180 g
Battery life	14.0 hours active
Telemetry channels	7

Parameter	Value
Data-link	2.4 GHz encrypted (paired only to Cell Halcón comms)
Charging	Inductive cradle, 90 minutes to full
Alert thresholds	Three-tier (caution / hold / abort)

The bracers issue a haptic pulse at three documented fatigue thresholds. Empirically, the **abort threshold** corresponds to a 73% probability of next-activation failure and is overridden by the asset only under formal commander's authorization. Override events: 4 logged (3 successful, 1 producing the August 2023 Echo Lock).

4.7 Anchor Sling

The **Anchor Sling** is a nylon-aramid harness used to redirect suspended objects in close quarters. Worn across the torso, the sling provides clip-points for batons and is constructed of **440 g** of woven mil-spec material rated to a static load of 1,200 kg. Standard configuration includes 6 baton clips, 2 ferro-rod ignition sleeves, and a single-point release.

4.8 Encrypted Comms: Yatiri-3 Burst-Radio

The **Yatiri-3** burst-radio handset is the standard PAMC Andean-cell comms unit:

Parameter	Value
Weight	460 g
Burst-transmission cycle	4.2 seconds
Range, line-of-sight	18 km
Range, via relay node	up to 240 km
Encryption	AES-256, Cooperative-issued keys, rotation every 96 hours
Battery	Lithium-polymer, 22-hour standby, 6-hour active

A backup **Yatiri-2** handset is cached at each operational safehouse.

4.9 Medical Kit

Standard medical loadout, kept in a dedicated thigh-pouch:

Item	Quantity	Notes
Calexin-7 (240 mg blister)	14 doses	Migraine prophylaxis; see §2.15
Hypertonic glucose gel	4 sachets	25 g fast-acting carbs each
Bofedal-3 electrolyte solution	2 × 740 mL	PAMC standard, see §2.14
Single-dose epinephrine pen	2	Standard 0.3 mg
Hyperbaric mask	1	Compact, for altitude emergencies

Item	Quantity	Notes
Stimulant blister (Cardex-2 analog)	6 doses	Restricted use, commander-authorization only
Hemostatic gauze	4 packs	Combat trauma standard
Tourniquet (CAT-7 variant)	2	One arm-rated, one leg-rated

The medical kit is inspected by Cpl. Vendrame before every deployment and after every operational return.

4.10 Climbing Kit

A reduced-volume mountaineering kit is integrated into the loadout for the asset's primary operational environment:

Item	Specification
Ice tools (×2)	Petzlec HK-2 , technical climbing rated
Crampons	Petzlec Lynx-A, semi-automatic
Harness	Camp Rapid-X, 280 g
Static rope	60 m × 8.6 mm
Ice screws	6 (lengths 13, 16, 19, 22 cm)
Carabiners	14 (8 locking, 6 non-locking)
Belay device	Camp Druid Pro
Headlamp	Faro-7 , 720 lumens, 11-hour battery
Snow stakes	2 × T-section aluminum

4.11 Vehicle: Quiltro V-12

Primary vehicle is an armored **Quiltro V-12** light utility, manufactured by **Industrias Ñuble** in Concepción under PAMC contract:

Parameter	Value
Mass	4,400 kg (4.4 t)
Drivetrain	4WD, dual-clutch automatic
Range	620 km (auxiliary tank engaged)
Top speed	162 km/h (governed)
Armor rating	NIJ Level III all-around, Level IV at A-pillars
Tires	Run-flat to 80 km at 50 km/h
Crew	4 + cargo

Parameter	Value
Vehicle ID	QV12-PAMC-019

The Quiltro is fitted with a concealed roof-rack mount for baton-cache deployment, an EM-shielded comms cabinet, and a 14-liter cold-storage compartment for Calixin-7 reserves.

4.12 Failsafe Equipment

Three classes of failsafe are standard:

- 1. **Dead-man's pulse beacon** — body-worn, transmits operative-alive signal every 240 seconds; absence of signal for ≥ 18 minutes triggers a Cell Halcón recovery protocol.
- 2. **Biometric pistol lock** — TX-9 trigger interlock keyed to the asset's palm geometry and pulse signature; lockout time 0.18 s.
- 3. **Concealed kill switches** — three hardware kill switches integrated into the Reset Bracers, Yatiri-3 handset, and Quiltro V-12 ignition, disabling each unit in the event of capture.

4.13 Loadout Variants

Three formal loadout variants are maintained:

Variant	Designation	Use	Mass
Standard	HALC-014-S	General Andean operations	18.4 kg
Urban	HALC-014-U	Cities, low-altitude	14.1 kg (no climbing kit, reduced batons x4)
High-altitude	HALC-014-H	Above 4,800 m	21.6 kg (added oxygen, doubled gloves)
Maritime	HALC-014-M	Coastal / port operations	16.7 kg (waterproof additions, no crampons)

4.14 Specialized Acquisitions and One-Off Equipment

Item	Description
Anchor Stone "Cerro-1"	Polished slate fragment recovered from the 2018 trigger site; carried as personal talisman in left chest pocket. Mass 32 g. No documented operational function.
Mnemonic Recorder Lente-9	CBMP-issued helmet camera with 8-hour buffer, used for post-action telemetry; deployed selectively.
EM-Suppression Cloak	Aluminized mylar-aramid weave; 1.2 kg; rated to attenuate 100 MHz–6 GHz emissions by 24–28 dB. Used 11 times since 2022.

4.15 Equipment Summary Matrix

Item	Manufacturer	Weight	Replacement (USD)
Andina-7 hardshell	Tejidos Cordillera	3.8 kg	\$4,200
Cordillera Mk-IV boots (pair)	Calzados Pirque	1.96 kg	\$620
Tungsten throwing batons (×6)	Talleres Magallanes	2.04 kg	\$5,640
Bersa-Nogales TX-9	Bersa-Nogales	0.72 kg	\$980
Reset Bracers (pair)	CBMP / Bauer-Cortez	0.36 kg	\$11,200
Anchor Sling	Tejidos Cordillera	0.44 kg	\$310
Yatiri-3 burst-radio	PAMC Signals	0.46 kg	\$2,800
Medical kit (full)	PAMC Medical	1.8 kg	\$440
Climbing kit (full)	Mixed	4.6 kg	\$3,100
Quiltro V-12 vehicle	Industrias Ñuble	4,400 kg	\$214,000
EM-Suppression Cloak	CBMP	1.2 kg	\$7,400

4.16 Historical and Retired Equipment

Equipment formally retired from the asset's loadout but retained in the **Cuartel Valparaíso Equipment Archive (CVEA)** for reference, training, and contingency reissue:

Item	Period in service	Reason for retirement
Andina-4 hardshell (predecessor uniform)	2019–2021	Replaced after Antofagasta thermal damage
Bersa-Vargas TS-7 sidearm	2019–2022	Trigger-pull fatigue; replaced with TX-9
Throwing batons "M-Series" (steel-only, 280 g)	2019–2020	Insufficient kinetic mass for Tail-Strike
Yatiri-1 burst-radio	2019–2023	Encryption deprecated; AES-128 sunset
Reset Bracer prototype A	2020–2021	Single-channel telemetry only
Quiltro V-9 vehicle	2019–2022	Damaged beyond repair, Mendoza 2022
Aluminized poncho (non-cloak variant)	2019–2022	Inferior EM-attenuation to current Cloak

The CVEA archive is consulted approximately twice per year by CBMP for comparative-evolution studies and once per training cycle by Lima Academy.

4.17 Quartermaster and Custody Chain

All operational equipment is signed in and out through the **Valparaíso Quartermaster's Log (VQL)**, currently maintained by **Sub-Comisario Eladia Munizaga-López** (PAMC civilian, tenure since 2019). VQL audits occur monthly. Lost-equipment incidents through 2025: **23 total**, of which 18 were recovered, 4 were declared total loss, and 1 (Baton B-3, Mendoza 2022) is unaccounted for and assumed in adversary possession.

4.18 Specialized Munitions Authorization

The asset is authorized to draw the following non-standard munitions from the Valparaíso armory upon commander's signature:

- **9×19 mm subsonic frangible** (Concession-grade): 60-round box
- **9×19 mm armor-piercing** (PAMC restricted): 30-round box, two-officer release
- **Smoke grenades**: M-18 equivalent, up to 4 per deployment
- **Flash-bang devices**: M-84 equivalent, up to 2 per deployment (note: must be deployed *behind* the asset to avoid acoustic disruption)

Munitions usage in 2024–2025: 14 subsonic rounds, 0 AP rounds, 6 smoke grenades, 1 flash-bang.

Section 5. Operational Tactics and Combat Doctrine

5.1 The Doctrine of Sequential Suspension

The asset's formal combat doctrine, the **Doctrine of Sequential Suspension (DSS)**, was articulated by the asset on **19 June 2022** in the aftermath of the Mendoza Pyrrhic engagement (§3.8). It is now PAMC-recognized doctrine for Class A-2 (Targeted) operatives and forms part of the standard curriculum at the Lima Operational Academy.

The doctrine rests on six core principles:

1. **Suspension is a decisive resource, not a continuous capability.** Treat each activation as ammunition.
2. **Sequence determines lethality.** A single suspension rarely wins an engagement; ordered chains of three define the doctrine.
3. **Acquisition is the vulnerability.** Protect the 1.2-second targeting window above all.
4. **Tail momentum is the payoff.** Plan terminal vectors before activating.
5. **Cooldown discipline is non-negotiable.** Operate as if the matrix is enforced by a third party.
6. **Withdraw before exhaustion.** Three activations and reset.

5.2 The Three-Two-One Rule

The asset's signature tactical heuristic, drilled into Cell Halcón operations and taught at Lima Academy:

- **Three** activations maximum per 60-second window
- **Two** active sightlines maintained at any moment
- **One** pre-anchored extraction route held in reserve

Violations of any element trigger an automatic call-off by the cell handler (Lt. Bauer-Cortez monitors via Reset Bracer telemetry). Documented violations: 7 since 2022, of which 5 produced operational

anomalies.

5.3 Tier-Based Engagement Protocols

PAMC Threat Tier definitions are detailed in §7.1. Engagement protocols differ as follows:

Threat tier	Asset's permitted engagement	Standing rules
Tier 1	Solo, full authority	No restrictions
Tier 2	Solo or cell, full authority	Comms check every 12 minutes
Tier 3	Cell required, lead authority	Three-Two-One enforced strictly
Tier 4	Cell required, mass-cooldown protocol	Calexin-7 dose pre-loaded; PAMC HQ on standby relay
Tier 5	Cell + auxiliary cell, Director's authorization required	Full pre-mission CBMP medical clearance, 96-hour rest preceding

The asset is **not authorized to engage Tier 5 threats solo under any condition.**

5.4 The Tail-Strike Technique

The **Tail-Strike** exploits Tail Momentum (§2.6) to deliver compounded impacts that cannot be achieved by linear projectile motion alone:

1. The asset throws a baton at a target.
2. Mid-flight, she suspends the baton, halting it in space.
3. While suspended, she physically repositions the baton (e.g., rotating it, advancing it along its path with a hand-push).
4. On release, the baton resumes its original momentum from its new position — striking with vector consistent with original throw but originating closer.

The result: a strike delivered at full thrown velocity but from an arbitrary, target-adjacent location. Adversary counter-prediction of trajectory becomes nearly impossible. Cumulative Tail-Strike applications logged through 2025: **312** (§2.17).

5.5 The Drop-Tune Manoeuvre

Drop-Tune is the standard activation pattern around the contact-exclusion rule (§2.8):

1. Asset releases an object she is holding.
2. After ~0.3 seconds, the contact-exclusion field clears.
3. Asset suspends the released object before it strikes anything.

The technique permits a meaningful one-second window between release and engagement target acquisition. The exact 0.3 s minimum has been internalized as muscle memory; the asset's Cell Halcón training videos use a metronome cue to instill the rhythm in new operatives.

5.6 The Suspended Field Technique

The **Suspended Field** uses the tertiary **Hold** ability (§2.11) to maintain multiple low-mass objects (typically batons) in mid-air simultaneously, creating a transient kinetic-cover zone:

- Maximum simultaneously held objects: **4** (matching available activations within a 60-s window minus a 1-suspension reserve)
- Hold duration per object: up to 4 s
- Field deployment time: ~12 seconds
- Field collapse: simultaneous release of all held objects with full Tail Momentum, producing a directed multi-baton strike

Cumulative Suspended Field deployments through 2025: **89**.

5.7 Mountain-Warfare Carryover Doctrine

The asset's Chilean Army background informs four standing carryover doctrines:

1. **Snow-Anchor Triage:** never advance past an anchor without verifying it. Carried over from rope-rescue protocols.
2. **Three-Point Contact:** always maintain three secure contacts (hand, foot, anchor) when traversing vertical terrain — applied to mark-placement scenarios.
3. **Altitude Acclimatization Protocol:** above 3,800 m, mandatory 48-hour acclimatization period unless emergency engagement.
4. **Whiteout Withdrawal:** below 12 m visibility, withdrawal is automatic regardless of mission priority.

5.8 Civilian Collateral Threshold

PAMC Standing Order 9 specifies a **zero-tolerance threshold below 80% confidence** that no civilian harm will occur from any planned activation. The asset is empowered to abort engagement at her sole discretion if civilian risk exceeds the threshold. Documented aborts on civilian-risk grounds since 2019: **31**.

5.9 Reconnaissance Sequencing

Standard pre-engagement reconnaissance for cell-led operations follows the **72-hour saturation protocol**:

- T-72h to T-48h: passive surveillance (Pajarera asset network if available)
- T-48h to T-24h: physical site survey by Sgt. Quispe-Linares
- T-24h to T-12h: electromagnetic survey by Lt. Bauer-Cortez
- T-12h to T-4h: medical and metabolic prep (Cpl. Vendrame)
- T-4h to T-0h: final cell briefing and equipment check

Solo deployments allow compression of this schedule to a 24-hour minimum, but not less.

5.10 The Four-Minute Window

The **Four-Minute Window** is the asset's standing extraction discipline: from initiation of withdrawal, the cell must reach the pre-anchored extraction route within **four minutes**. The figure derives from observed adversary response times during the Mendoza engagement and has held as a planning constant since 2022.

5.11 The Twelve Combat Directives

The asset's published standing combat directives (PAMC Internal Bulletin 22-19):

1. Acquire targets only when extraction is mapped.
2. Never activate without verifying the contact-exclusion radius.
3. Hold one activation in reserve until extraction.
4. Cooldown discipline overrides tactical opportunity.
5. Civilian risk overrides cooldown discipline.
6. Engage Tier 4 only with two extraction routes verified.
7. Never self-suspend while gripping a weapon.
8. Never engage above 4,800 m without acclimatization.
9. Never engage below 12 m visibility.
10. Withdraw at 80% targeting fatigue (Reset Bracer abort threshold).
11. Always release a captured Calixin-7 supply to the medic before engagement.
12. Never break radio silence on the Yatiri-3 burst comms during targeting.

5.12 Documented Doctrine Departures

Date	Incident	Departure	Outcome
17 May 2021	Antofagasta	Directive 9 violated (low visibility)	Successful but injury sustained
4 Jun 2022	Mendoza	Directive 10 violated (continued past fatigue threshold)	Pyrrhic (civilian casualties)
14 Aug 2023	Salta	Directive 6 violated (engaged Tier 4 with single route)	Catastrophic

Each departure is referenced in formal after-action review materials and is taught at Lima Academy as a doctrinal case study.

Section 6. Allies, Networks, and Known Affiliations

6.1 The Pan-American Metahuman Cooperative (PAMC)

Parent organization: the **Pan-American Metahuman Cooperative (PAMC)**, founded **22 May 2014** through the Treaty of Quito by the governments of nine Western Hemisphere states. Headquartered at **Edificio Marasol**, Avenida Salaverry 1840, Lima.

- Founding signatories: Chile, Peru, Bolivia, Argentina, Colombia, Ecuador, Costa Rica, Uruguay, Canada (observer)
- Current member states: 12 (added Mexico 2017, Panama 2019, Paraguay 2023)
- Registered operatives: 217 (as of 1 January 2026)
- Director-General: **María-Elena Espósito-Tagle** (Argentine; term 2022–2027)
- Budget (FY 2026): USD \$382 million across all directorates

The Cooperative operates four directorates: Operations, Behavioral Health, Logistics, and the Cooperative Bureau of Metahuman Physics (CBMP). SLATEFALL is administratively assigned to the **Andean Operations Bureau** under the Operations Directorate.

6.2 Cell Halcón

Cell Halcón is the four-operative tactical unit under SLATEFALL's command. Composition was finalized 11 October 2022:

Member	Role	Status	Joined cell
SLATEFALL (Inez Calvache-Renström)	Lead Officer	Metahuman A-2	1 Aug 2022
Lt. Solène Bauer-Cortez	Technical Specialist	Metahuman C-1 (EM sensing)	11 Oct 2022
Sgt. Mateo Quispe-Linares	Tracker / Recon	Civilian	11 Oct 2022
Cpl. Raúl Vendrame	Field Medic	Civilian	18 Jan 2023

The cell is rated for **Tier 4** engagements and has logged **47 operations** as a complete unit through 31 December 2025. Cell mortality: 0. Cell injury rate: 4 (non-fatal) across all operations.

6.3 Cell Halcón Member Profiles

Lt. Solène Bauer-Cortez

- Age 31, born Geneva, Switzerland; naturalized Argentine citizen 2017
- PAMC tenure 5 years
- Metahuman classification: **Class C-1 (Electromagnetic Sensing)** — perceives EM emissions from 30 Hz to 6 GHz within 80 m
- Engineering education: ETH Zürich, MSc Electrical Engineering 2018
- Designed the Reset Bracers (§4.6)
- Languages: French, German, Spanish, English
- Reports directly to SLATEFALL for tactical matters; reports to Director Salvatierra for technical R&D

Sgt. Mateo Quispe-Linares

- Age 38, born La Paz, Bolivia
- Civilian (no metahuman classification)
- Prior service: Bolivian Special Forces, **Regimiento Bolívar**, 14 years
- Specialty: high-altitude tracking, alpine reconnaissance
- Languages: Spanish, Aymara, Quechua, English
- Joined PAMC 2019; rotated through three cells before Cell Halcón

Cpl. Raúl Vendrame

- Age 29, born Mendoza, Argentina
- Civilian
- Prior service: Argentine Army medical corps, 6 years
- Specialty: trauma surgery, altitude medicine, metahuman physiology (PAMC-certified 2023)
- Languages: Spanish, English, Portuguese (working)
- Primary administrator of the asset's Calixin-7 regimen during operations

6.4 Senior PAMC Allies

Director Aída Salvatierra-Bunge

- Age 58, born Buenos Aires
- Director, PAMC Operations Directorate, Lima
- PAMC career since 2015; prior career as Argentine Federal Intelligence officer
- The asset's direct chain-of-command superior since 2022
- Authorized to override all engagement decisions except Director-General-level orders

Dr. León Marrón-Saavedra

- Age 64, born Mérida, Venezuela
- Senior scientist, CBMP
- Co-author of the Calvache–Marrón Equations
- Conducts annual scientific re-evaluations of the asset's power (§2.19)

Dr. Maite Borghi-Castelnuovo

- Age 47, born Lima
- Senior psychologist, PAMC Behavioral Health
- The asset's continuing mental-health practitioner since 2020

6.5 External Civilian Allies

Dr. Catalina Renström-Verdugo (spouse)

- Civilian, no metahuman classification
- Pediatric oncologist, Clínica Las Condes, Santiago
- PAMC Civilian Liaison Clearance Level 3 (granted January 2024)
- Occasional consultant on pediatric-related civilian rescue operations (2 documented advisory cases)

Comandante Ezequiel Ovalle-Garrido

- Age 51, Chilean Army active duty
- Liaison officer between PAMC and Chilean Army Andean Brigade
- Long-standing professional relationship with the asset (pre-PAMC, dating to 2014)
- Authorized to coordinate Chilean Army support for PAMC operations in Chilean territory

"Pajarera" / Tessa Roehl

- Age 43, dual German-Bolivian citizen
- Civilian intelligence asset based in La Paz
- Operates a small import-export consultancy as cover
- Contact frequency: approximately monthly
- Compensated through Cooperative discretionary fund

Hermana Asunta Mendieta

- Age 67, Bolivian, member of the Order of the Sisters of Mercy
- Non-combatant safehouse keeper, Cuzco (PAMC Safehouse C-3, §8.4)
- Tenure with PAMC: since 2017
- Provides medical and shelter services during transit operations

6.6 Former Affiliations

- **Chilean Army** — formal affiliation ended 28 February 2019, but informal channel remains through Comandante Ovalle-Garrido
- **Escuela Militar Bernardo O'Higgins** — alumna; occasional guest lecturer when on leave (2 lectures since 2022)
- **CEM Río Blanco** — former instructor; consulted by current cadre on at least 4 occasions since 2019
- **OEA-MAR** — formal affiliation lapsed; informal coordination remains

6.7 Tolerated-But-Hostile Relationships

- **PAMC Inspector General Magnus Sletten** (the "Underwriter" — see §7.6) — adversarial bureaucratic figure within the same parent organization. Formal correspondence channel maintained; no in-person meetings since 2024.
- **Operative MARTILLO / Joaquín Hidalgo-Yáñez** — PAMC Class A-1 operative, Caracas Cell. Documented professional friction since 2023; coordination only when mandated by command.

6.8 Allies Roster Summary

Name / Codename	Role	Tenure with asset	Trust rating (1–5)
Solène Bauer-Cortez	Cell technical lead	3 yrs	5
Mateo Quispe-Linares	Cell tracker	3 yrs	5
Raúl Vendrame	Cell medic	3 yrs	5
Aída Salvatierra-Bunge	Operations Director	4 yrs	4
Léon Marrón-Saavedra	CBMP physicist	7 yrs	4
Maite Borghi-Castelnuovo	Psychologist	6 yrs	5

Name / Codename	Role	Tenure with asset	Trust rating (1–5)
Catalina Renström-Verdugo	Spouse	4 yrs (relationship); married 2 yrs	5
Ezequiel Ovalle-Garrido	Army liaison	12 yrs	4
Tessa Roehl ("Pajarera")	Civilian intel	3 yrs	3
Asunta Mendieta	Safehouse keeper	4 yrs	4

6.9 Operational Auxiliaries and Reserve Contacts

In addition to the principal allies enumerated above, the asset maintains a network of **auxiliaries** — personnel cleared for limited operational interaction without full PAMC affiliation. As of revision date, 14 auxiliaries are listed in the asset's Cooperative-approved contact roster:

Auxiliary	Role	Location	Affiliation
Cap. Diego Maluenda-Riffo	Chilean Carabineros liaison	Valparaíso	Chilean national police
"Carrera" (true name redacted)	Black-market munitions supplier	Iquique	Independent
Profa. Aliéndez de la Vega	UCh historian, archival access	Santiago	Civilian academic
Dr. Bartolomé Wirth-Yáñez	Veterinarian, animal-rescue extractions	Mendoza	Civilian
"Plata Vieja"	Currency runner, micro-cache top-ups	La Paz	Independent
Insp. Hilario Pacheco-Solar	Argentine federal police, intel exchange	Salta	Argentine state
"Sastre" (true name redacted)	Forger of secondary identity documents	Cuzco	Independent
Dr. Verónica Hagedorn-Mesa	Pediatric trauma surgeon, civilian-casualty triage	Antofagasta	Civilian
Capt. (ret.) Fermín Calderón-Brun	Former CEM cadre, mountain extraction	Coyhaique	Civilian, ex-military
"Tórtola"	Communications relay operator	Trelew	Civilian
Lic. Marta Spiess-Quiroga	Lawyer, civilian-affairs counsel	Lima	Civilian
Comdt. Erlind Halvorsen	Norwegian Coast Guard liaison (anti-Concession)	Tromsø	Foreign government

Auxiliary	Role	Location	Affiliation
"Ñandú" (true name redacted)	Drone operator, surveillance contracts	Bariloche	Independent
Sra. Inocencia Lemus-Vargas	Cook and housekeeper, S-1 safehouse	Punta Arenas	Civilian

Auxiliaries are recompensed via Cooperative discretionary funds at standardized rates (USD \$400–\$2,400 per engagement, depending on risk class). All payments are logged and audited annually by the PAMC Inspector General's office.

6.10 Inter-Cooperative Liaisons

Formal liaison relationships are maintained with three sister organizations:

- **Nordic Metahuman Operations Cell (NMOC)** — joint training rotations every 18 months
- **European Metahuman Reserve (EUMR)** — case-data exchange, four reciprocal briefings per year
- **East Asian Metahuman Compact (EAMC)** — observer status, biennial conference attendance

The asset attended the **2024 Pan-Cooperative Summit** in Lisbon as one of three PAMC representatives, presenting a redacted briefing on the Doctrine of Sequential Suspension.

Section 7. Adversaries and Documented Threats

7.1 PAMC Threat Tier Classification System

The Cooperative classifies metahuman and organized-paramilitary threats into five tiers under the **NMOC-derived Threat Tier Schedule**, ratified 11 March 2017 and most recently amended Q1 2025. Tier assignment is governed by the following quantitative criteria:

Tier	Power output ceiling	Casualty potential (unmitigated)	Engagement authority required
Tier 1	< 0.5 kJ kinetic equivalent	≤ 1 civilian	Field operative, solo
Tier 2	0.5–25 kJ	2–10 civilians	Field officer, solo or cell
Tier 3	25–500 kJ	11–100 civilians	Lead officer, cell required
Tier 4	500 kJ – 50 MJ	101–5,000 civilians	Lead officer + auxiliary on standby
Tier 5	> 50 MJ or strategic-scale	> 5,000 civilians	Director-General authorization

Tier ratings are reassessed annually or upon documented capability change. The asset has personally engaged threats up to Tier 4 (Tropopausa, Quebrantadero); no Tier 5 engagement is presently

authorized.

7.2 Named Adversary Roster — Overview

The asset's documented adversary roster comprises **seven principal named threats** and three networked organizations. The full roster is reviewed quarterly by the Andean Operations Bureau.

Codename	Civilian ID	Tier	Status	First contact
Tropopausa	Eliseo Mardones-Vodanovic	4	Active, contained 2024	3 Jun 2022
El Sereno	Unknown	3	Active, at-large	11 Apr 2023
Quebrantadero	Wilma Sandoval-Iqui	4	Active, escaped 2023	14 Aug 2023
Cabrestante	Pablo Eyzaguirre-Pinto	3	Captured 2025	18 Sep 2024
Underwriter	Magnus Sletten	2 (bureaucratic)	Active, in-system	9 Feb 2024
Mar Inverso	Unknown	5 (projected)	Active, unconfirmed	22 Dec 2025
Iron Tributary	Multiple	3 (organizational)	Active	7 Mar 2024

7.3 Tropopausa / Eliseo Mardones-Vodanovic

Codename: Tropopausa **Civilian identity:** Eliseo Mardones-Vodanovic, born 1981, San Bernardo, Chile **Profession (cover):** atmospheric scientist; formerly affiliated with the **Centro de Estudios Climáticos de Valparaíso (CECV)** **Tier:** 4

Powers: atmospheric pressure manipulation within a 40-meter radius. Capable of inducing local pressure differentials up to **±0.62 atm** in a 4-second interval, producing acute barotrauma in unprotected personnel and disabling small aircraft engines. He cannot create wind ex nihilo but can redistribute existing pressure gradients with high precision.

Documented limits:

- Field-creation cooldown: 18 seconds between activations
- Maximum sustained field duration: 22 seconds before metabolic collapse
- Vulnerable to acoustic disruption above 140 dB
- Cannot operate above 5,800 m altitude (atmospheric thinning disables binding)
- Class A-2 (Targeted) — same scientific class as SLATEFALL, but on a different operational axis

Threat profile vs. SLATEFALL: Tropopausa's pressure fields can interfere with SLATEFALL's targeting acquisition by introducing tinnitus-grade interior-ear disturbance. Acoustic shock from his pressure pulses has produced one documented targeting interruption (Mendoza, 2022).

Engagement history: 4 documented engagements through 2025 (Mendoza 2022, Antofagasta 2023, Iquique 2023, Coquimbo 2024). Currently contained at the **PAMC Restraint Facility, Iquique** (since November 2024).

7.4 El Sereno

Codename: El Sereno **Civilian identity:** Unknown; suspected male, Andean ancestry, age 30–45 **Tier:** 3

Powers: nocturnal sensory enhancement (multi-spectral, including thermal IR to 14 μm) and a documented capacity for kinetic dampening of his own footfalls and respiration. Approaches targets in absolute silence; documented sound floor during approach: -12 dB SPL (below ambient).

Documented limits:

- All capabilities suppressed by direct sunlight or visible-spectrum illumination above **1,200 lux**
- Documented hearing impairment in left ear (left-side targeting penalty)
- Operates exclusively between local dusk and 1.5 hours before local dawn

Threat profile vs. SLATEFALL: El Sereno is the asset's most dangerous nocturnal threat because his stealth approaches degrade SLATEFALL's targeting acquisition (which depends on visual perception). Drift Read provides partial counter-defense within 3 m.

Engagement history: 3 confirmed engagements (Iquique 2023, Antofagasta 2024, Arica 2025); all inconclusive. He is the longest-standing at-large adversary on the asset's roster.

7.5 Quebrantadero / Wilma Sandoval-Iqui

Codename: Quebrantadero **Civilian identity:** Wilma Sandoval-Iqui, born 1979, Oruro, Bolivia **Profession (cover):** orthopedic surgeon (revoked license 2022) **Tier:** 4

Powers: induces controlled bone-density fluctuation in mammalian subjects within a 14-meter radius by line-of-sight contact. Effects range from temporary fragility (osteopenia-equivalent for ~ 6 hours) to acute fracture cascades.

Documented limits:

- Cannot affect non-mammalian biology
- Targeting requires 2.4-second focus, with audible vocalization
- Counterable by lead-lined shielding (≥ 6 mm thickness)
- Effect non-cumulative; subsequent re-targeting within 30 minutes produces only 12% additional density change

Threat profile vs. SLATEFALL: Quebrantadero's effect is targeted at biological tissue, not inanimate matter — meaning SLATEFALL's primary kinetic implements are unaffected, but the asset herself is highly vulnerable. The 14 August 2023 Salta engagement (§3.10) produced 11 civilian casualties from a Quebrantadero pulse the asset failed to interdict.

Engagement history: 2 documented engagements (Salta 2023, Cuzco 2024). Currently at-large.

7.6 Underwriter / Magnus Sletten

Codename: Underwriter (asset-side designation; he holds no internal codename) **Civilian identity:** Magnus Sletten, age 58, Norwegian-Argentine dual citizen **Position:** PAMC Inspector General (Administrative Compliance), Lima HQ **Tier:** 2 (bureaucratic threat assessment; no kinetic capability)

Description: Inspector General Sletten is classified an adversary not because of metahuman capability but because of sustained adversarial bureaucratic action against the asset's operational profile. Between 2024 and 2026 he initiated **eleven formal audits** of SLATEFALL operations, three of which sought disciplinary referral; all three were dismissed. His professional position renders him institutionally protected; engagement protocols are therefore non-kinetic and procedural only.

Threat profile vs. SLATEFALL: the threat is operational restriction, not physical harm. Case 2025-0019 (§9.8) was a counter-audit defensive engagement, terminating in audit reversal by the Director-General.

7.7 Cabrestante / Pablo Eyzaguirre-Pinto

Codename: Cabrestante **Civilian identity:** Pablo Eyzaguirre-Pinto, born 1975, Calama, Chile **Profession (cover):** mining-machinery engineer **Tier:** 3

Powers: none innate; classified as a metahuman-adjacent adversary by virtue of his cybernetic augmentation. His augments include:

- Reinforced cervical and thoracic skeleton (titanium-composite, +18 kg total mass)
- Hydraulic right arm rated to 4.2-ton lift, 1.4-ton sustained
- Optical implant providing 8× zoom and thermal vision
- Subcutaneous power-cell array, 14-hour duty cycle

Documented limits:

- Augments fail catastrophically below −12 °C ambient
- EMP exposure above 0.2 T disables optical and hydraulic systems for 4–12 hours
- Augment maintenance requires monthly servicing at a specialized facility (one suspected facility identified, Potosí, 2024)

Threat profile vs. SLATEFALL: Cabrestante's hydraulic arm exceeds the 240 kg mass ceiling for direct Inertial Suspension, but separate suspended-component attacks (e.g., suspending his arm at the elbow joint) have proven effective. Combat record vs. SLATEFALL: 1 engagement (Potosí, September 2025, Case 2025-0156, §9.9), captured.

7.8 Iron Tributary

Designation: Iron Tributary **Type:** organized paramilitary cell, Caucasus-trained, 22 confirmed members **Tier:** 3 (organizational; mixed individual capability) **Operational territory:** northern Argentina, Bolivia, southern Peru **Funding:** suspected illicit mining revenue, USD ~\$28 million annually

Iron Tributary is a paramilitary group of mixed-discipline operatives, including at least 3 metahuman members (1 Tier 2, 2 Tier 1). The cell has clashed with PAMC operations 7 times since 2022. SLATEFALL has engaged the cell on 2 occasions (Tarapacá 2024, Cuzco 2025), with successful outcomes.

7.9 The Concession

Designation: The Concession **Type:** corporate-paramilitary syndicate **Membership:** 62 confirmed members, estimated up to 110 total **Headquarters (suspected):** Industrial Park "El Tronco," Antofagasta Province **Tier:** Tier 3 organizational; individual operatives range Tier 1–3

The Concession is a private-sector security syndicate offering metahuman-augmented services to the mining and energy industries of the Southern Cone. It has engaged PAMC operations 14 times since 2022; SLATEFALL has personally engaged Concession operatives 5 times (Tarapacá raid Case 2024-0034 being the most significant, §9.6).

Key Concession personnel known to the asset:

- **Hugo Lacassagne-Esteban** — head of operations, age 49, former Argentine Federal Police
- **"Cataract" / Helene Drovac** — optical-disruption metahuman, Tier 3, Concession contract operative
- **Yannik Brizuela** — financial controller, civilian

7.10 Mar Inverso (Projected Tier 5)

Codename: Mar Inverso **Civilian identity:** Unknown **Tier:** projected 5; unconfirmed **First indication:** 22 December 2025, intelligence intercept (source: Pajarera)

Mar Inverso is the asset's most pressing emergent threat. Intelligence intercepted in late 2025 references a metahuman with **inertial inversion capability** — projected to be the operational inverse of SLATEFALL's power, capable of multiplying the inertia of target objects rather than nullifying it. No direct engagement has occurred; physical evidence is limited to two unexplained vehicle incidents (Mendoza port, August 2025; La Paz suburb, November 2025) in which heavy vehicles exhibited momentary mass-spikes inconsistent with conservation principles.

If confirmed, Mar Inverso would represent the only known operational counter-symmetric threat to SLATEFALL's primary power. PAMC has elevated the file to **Andean Operations Bureau Priority 1** as of January 2026. The asset is not currently authorized to engage Mar Inverso solo.

7.11 The Audiencia

Designation: The Audiencia **Type:** bureaucratic-adversary cluster within PAMC oversight bodies **Membership:** estimated 7–12 senior administrators (4 confirmed) **Tier:** 2 (bureaucratic, non-kinetic)

The Audiencia is a faction within PAMC's senior administration that opposes the current operational profile of Class A-2 assets, including SLATEFALL. The Underwriter (§7.6) is associated but not formally a member. The Audiencia's documented actions include three failed motions to reclassify SLATEFALL to a non-combatant role (2024 ×1, 2025 ×2).

7.12 Adversary Engagement Frequency Statistics

Cumulative engagement statistics across all named threats:

Adversary	Engagements	Wins (PAMC)	Losses	Inconclusive
Tropopausa	4	3	0	1
El Sereno	3	0	0	3

Adversary	Engagements	Wins (PAMC)	Losses	Inconclusive
Quebrantadero	2	0	1	1
Cabrestante	1	1	0	0
Underwriter	(bureaucratic only)	3	0	0
Iron Tributary	2	2	0	0
The Concession	5	4	0	1
Mar Inverso	0	—	—	—

7.13 Emergent Threat Forecast (2026–2027)

The Andean Operations Bureau projects the following emergent threats over the next 18 months:

1. **Mar Inverso confirmation and engagement** — high probability within 2026.
2. **Quebrantadero re-emergence in northern Chile** — moderate probability; medical-supply intercepts suggest she remains in the region.
3. **El Sereno geographic expansion** — sighting pattern suggests southward expansion into Atacama.
4. **The Concession diversification** — financial intelligence suggests planned expansion into Andean lithium operations; likely Cell Halcón engagement within 12 months.
5. **A new bureaucratic challenge** from the Audiencia following the 2026 PAMC Annual Review.

7.14 Minor Adversaries and Watch-List Entities

The asset's secondary threat roster — entities flagged for monitoring but not yet considered primary engagement risks — comprises eight Watch-List entries (W-001 through W-008):

ID	Designation	Capability summary	Tier	Status
W-001	" Calamina " / Patricio Rojas-Salinas	Sheet-metal manipulation, ≤ 6 mm thickness	2	Inactive since 2024
W-002	" Higrómetro " (identity unknown)	Hyper-localized humidity manipulation	1	At-large, low-priority
W-003	" Carbonero " / Iván Belalcázar-Yepes	Carbon-soot generation, asphyxiation risk	2	Custody (CBMP, Lima)
W-004	" La Pastora " / Yulia Martsenyuk	Animal-herd influence, livestock-only	1	Inactive
W-005	" Tornillero " / Davor Krznarić	Mechanical-fastener manipulation, ferrous metals	2	Active, Argentina
W-006	" Bocatoma " (identity unknown)	Small-stream redirection, ≤ 50 L/s	1	At-large, Patagonia
W-007	" Chasqui " / Eduardo Pizarro-Lanata	Localized footspeed boost, +35%, 90 s window	2	Active, Cuzco region

ID	Designation	Capability summary	Tier	Status
W-008	" Plomo Frío " (identity unknown)	Cooling of small ferrous masses to −60 °C	2	At-large, Bolivia

The Watch List is reviewed quarterly by Cell Halcón technical specialist Lt. Bauer-Cortez. Promotion to the primary adversary roster requires either three documented engagements or a single incident producing more than 5 civilian casualties.

7.15 Defunct or Neutralized Adversaries (Historical)

The following former adversaries are no longer active or are confirmed deceased and are retained for historical reference only:

- "**Bambolla**" / **Ezekiel Hartog-Maan** — explosive-fabrication metahuman; killed in own accidental detonation, 2022
- "**Reverso**" / **Anabella Trumler** — minor mirror-image projector; defected to PAMC custody 2023, currently civilian
- "**El Funicular**" (identity unconfirmed) — kinetic anchor-line manipulator; suspected deceased in 2024 avalanche
- "**Quilatera**" / **Hortencia Pavón-Vasconcelos** — gemstone-resonance metahuman; convicted civilian fraud, no metahuman activity since 2021

7.16 Threat Assessment Methodology Notes

The asset's adversary roster is compiled using the **NMOC-Lima Hybrid Scoring Matrix (NLHSM)**, which weights:

- Power output (40%)
- Operational sophistication (25%)
- Networked support (15%)
- Civilian-collateral propensity (15%)
- Anti-SLATEFALL countermeasure capability (5%)

Quarterly reassessment is performed by Lt. Bauer-Cortez with CBMP review.

Section 8. Known Bases, Safehouses, and Operational Territory

8.1 Primary Base: Cuartel Valparaíso

The asset's primary operational base is **Cuartel Valparaíso**, designated PAMC South Cone Headquarters. The facility occupies a decommissioned Chilean Navy fuel depot at **Avenida Altamirano 1842, Valparaíso**, repurposed for PAMC use in 2017.

Parameter	Value
Total floor area	4,820 m²
Personnel capacity	38 (residential), 142 (working)

Parameter	Value
Vehicle bay	12 spaces (including 1 armored bay for the Quiltro V-12)
Comms array	1 primary Yatiri relay node, 2 backup satellite uplinks
Helipad	1 (Bell-415 compatible)
Subsurface levels	2 (storage, CBMP regional micro-lab)
Average daily population	47 personnel
Operational security level	PAMC Restricted-3

Cuartel Valparaíso houses Cell Halcón's principal quarters, the asset's personal residence wing (44 m² apartment, Block C), and the regional CBMP medical office. The facility is hardened against EMP exposure and EM eavesdropping.

8.2 Secondary Bases

Base	Location	Function
Edificio Marasol	Avenida Salaverry 1840, Lima	PAMC Headquarters; the asset rotates through quarterly for senior briefings
Oficina Tarija	Calle Bolívar 412, La Paz	Field office, Bolivian altiplano operations
Annex Trelew	Industrial Park 7, Trelew, Argentina	Logistics-only annex, southern operations

8.3 Operational Safehouses

The asset has access to five formally maintained safehouses within her operational territory:

Safehouse	Location	Type	Status
S-1	Punta Arenas, Magallanes	Coastal, residential cover	Active
S-2	Mendoza, Argentina	Urban apartment	Active
S-3	Iquique, Tarapacá	Maritime, port district	Active
S-4	Antofagasta, Antofagasta	Industrial cover	Active (rebuilt 2024 after damage)
S-5	Cuzco, Cusco Region, Peru	Religious-order cover (Hermana Mendieta, §6.5)	Active

Each safehouse maintains a **standardized provisioning standard**:

- 30-day rations for 4 personnel
- Yatiri-2 backup comms unit
- 6 baton-cache replacement set

- 2 Calixin-7 spare blister packs
- 1 backup PAMC-issued identity package (Norwegian, Argentine, Bolivian alternates)
- 2,000 USD-equivalent local currency in mixed denominations

Safehouse provisioning is audited semi-annually by Cpl. Vendrame on behalf of the cell.

8.4 Micro-Cache Network

The asset maintains a network of **19 micro-caches** distributed across the Andean cordillera:

- **9 caches** in Chile (Atacama 3, central Chile 4, Patagonia 2)
- **4 caches** in Argentina (Mendoza Province 2, Salta 1, Neuquén 1)
- **3 caches** in Bolivia (La Paz 1, Potosí 1, Oruro 1)
- **2 caches** in Peru (Cuzco 1, Arequipa 1)
- **1 cache** in Ecuador (Cuenca)

Standard cache contents (PAMC Logistics Form MC-04):

Item	Quantity
Tungsten throwing batons (replacement)	2
Calixin-7 dose	4 blisters (8 doses)
Bofedal-3 electrolyte concentrate	4 packets
Trauma medical kit (compact)	1
Local currency	\$400 equivalent
Yatiri-2 backup handset	1
Identity package	1

Cache locations are rotated every 36 months; current placements were finalized in 2024 and will be reviewed in 2027.

8.5 Operational Territory

The asset's authorized operational territory spans **approximately 6.8 million km²**, encompassing the western and southern Andean states:

- **Chile** — full territory, including offshore islands except Easter Island (restricted)
- **Bolivia** — full territory
- **Peru** — full territory south of 10°S latitude
- **Argentina** — provinces of Mendoza, Salta, Jujuy, Neuquén, Río Negro, and Chubut
- **Ecuador** — limited to provinces of Azuay, Cañar, and Loja
- **Paraguay** — observer territory; no engagements authorized

Beyond this territory, the asset operates only under Director-General authorization. Two extra-territorial deployments have been authorized since 2019 (Quito 2021, Lima Pan-American Conference 2024).

8.6 Restricted and Forbidden Zones

Three zones within the asset's operational territory are formally restricted:

Zone	Reason	Status
Cerro Castillo exclusion ring (1.6 km radius around the 2018 trigger site)	Psychological; CBMP recommendation since 2020	The asset has not entered since the 2018 incident
Salar de Uyuni Sector 4 (Bolivia)	Strategic; ongoing classified PAMC investigation since 2023	Limited access, prior coordination required
Easter Island	Jurisdictional; pending bilateral agreement	No PAMC operations authorized

8.7 Communications Infrastructure

The asset's operational territory is served by **8 burst-relay nodes** within the PAMC Yatiri network:

- 1. Valparaíso (primary)
- 2. Antofagasta
- 3. Arica
- 4. La Paz
- 5. Cuzco
- 6. Salta
- 7. Mendoza
- 8. Punta Arenas

Node redundancy: each cell-led operation requires line-of-comms to at least 2 nodes. Operations outside relay coverage require satellite uplink and Director's authorization.

8.8 Territorial Agreements with Allied Operators

The asset's territory overlaps with several PAMC and partner-organization operators. Formal coordination agreements govern these overlaps:

- **Cell Cóndor (Mendoza-based)** — primary operator for Argentine Mendoza and Neuquén; SLATEFALL coordinates entry
- **Operative MIRADOR (Atacama)** — senior PAMC operative, the asset's former trainer
- **Bolivian Servicio de Operaciones Especiales (SOE)** — bilateral cooperation framework, La Paz Annex 2023
- **Peruvian Dirección de Asuntos Metahumanos (DAM)** — observation-only sharing for Cuzco operations

Section 9. Case Files: Documented Engagements and Incidents

9.1 Case File Format

All cases are recorded under a uniform PAMC format:

- **Case number** (year-sequence)

- **Date** (start–end)
- **Location** (city, province, country)
- **Classification** (open / classified / declassified)
- **Outcome** (Successful / Partial / Inconclusive / Pyrrhic / Catastrophic)
- **PAMC casualties** (operative, civilian)
- **Adversary casualties** (captured, killed, escaped)
- **Post-action notes**

9.2 Case 2019-0017: Probationary Assessment (Atacama)

- **Case number:** 2019-0017
- **Date:** 22–24 November 2019
- **Location:** Atacama Desert, near Tocopilla, Antofagasta Region, Chile
- **Classification:** Declassified Q4 2020
- **Outcome:** Successful
- **PAMC casualties:** none
- **Adversary casualties:** 2 captured (small-arms smugglers, civilian)
- **Notes:** First independent operation following probationary period. Demonstrated initial mass-ceiling stability at 200 kg. Operation concluded ahead of schedule (52 hours actual vs. 72 planned).

9.3 Case 2020-0048: Antofagasta Containment

- **Case number:** 2020-0048
- **Date:** 17–19 May 2021 (delayed action arising from December 2020 incident)
- **Location:** Port District, Antofagasta, Chile
- **Classification:** Declassified Q3 2021 (Report PAMC-PR-2021-006)
- **Outcome:** Successful, with operative injury
- **PAMC casualties:** 1 (asset, 14-day medical leave)
- **Adversary casualties:** 4 captured, 1 escaped
- **Notes:** Drift Read first manifestation (§3.7). The escaped adversary subsequently identified as a low-tier Concession affiliate. First post-action evaluation prescribed Calixin-7.

9.4 Case 2022-0123: Tropopausa First Engagement (Mendoza)

- **Case number:** 2022-0123
- **Date:** 3–5 June 2022
- **Location:** Mendoza Free Trade Zone, Argentina
- **Classification:** Classified (Tier-2)
- **Outcome:** Pyrrhic
- **PAMC casualties:** 1 operative KIA (Cpl. Adolfo Yáñez-Ríos, Cell Cóndor); 3 civilians killed
- **Adversary casualties:** Tropopausa contained but escaped after 8 hours
- **Notes:** Engagement triggered formal articulation of the Doctrine of Sequential Suspension. Asset suffered ear barotrauma from pressure pulse; recovered without permanent damage. Case led to revised tactical protocols against pressure manipulators.

9.5 Case 2023-0067: El Sereno Pursuit (Iquique)

- **Case number:** 2023-0067
- **Date:** 11–14 April 2023
- **Location:** Iquique, Tarapacá Region, Chile
- **Classification:** Open
- **Outcome:** Inconclusive
- **PAMC casualties:** none
- **Adversary casualties:** none (El Sereno disengaged at dawn)
- **Notes:** First contact with El Sereno. Demonstrated the limitations of nocturnal engagement profiles. Recommended development of high-lux operation kit; partially implemented 2024.

9.6 Case 2024-0034: Tarapacá Concession Raid

- **Case number:** 2024-0034
- **Date:** 17–18 February 2024
- **Location:** Tarapacá Region, Chile (compound near Pozo Almonte)
- **Classification:** Declassified Q1 2024 (Report PAMC-PR-2024-002)
- **Outcome:** Successful
- **PAMC casualties:** 1 wounded (Sgt. Quispe-Linares, leg laceration, 8-day recovery)
- **Adversary casualties:** 14 Concession affiliates captured; 2 killed in firefight; 1 metahuman ("Cataract") escaped
- **Notes:** Major operational success demonstrating Cell Halcón's full capability. The asset deployed Suspended Field for the first time in combat. Detained personnel led to seizure of USD \$4.2 million in Concession assets.

9.7 Case 2024-0202: Quebrantadero Showdown (Salta)

- **Case number:** 2024-0202
- **Date:** 14 August 2023
- **Location:** Salta city center, Salta Province, Argentina
- **Classification:** Classified (Tier-3); inquiry concluded March 2024
- **Outcome:** Catastrophic
- **PAMC casualties:** 0 operative; **11 civilian deaths**, 27 civilian injuries
- **Adversary casualties:** Quebrantadero escaped
- **Notes:** Second Echo Lock event during the engagement (§3.10). Quebrantadero's pulse struck a civilian-density urban target before SLATEFALL could interdict. Formal inquiry recommended three protocol changes (all adopted). Case remains the asset's most prominent operational failure and is a featured case study at Lima Academy.

9.8 Case 2025-0019: Audiencia Counter-Audit (Lima)

- **Case number:** 2025-0019
- **Date:** 22 January – 18 February 2025
- **Location:** PAMC HQ, Lima
- **Classification:** Sealed (internal proceeding)
- **Outcome:** Successful (audit reversed)
- **PAMC casualties:** none
- **Adversary casualties:** Underwriter referral dismissed; one Audiencia member reassigned

- **Notes:** Procedural defensive action; included formal testimony by SLATEFALL on 4 February 2025. Director-General Espósito-Tagle issued the final ruling.

9.9 Case 2025-0156: Cabrestante Neutralization (Potosí)

- **Case number:** 2025-0156
- **Date:** 18–22 September 2025
- **Location:** Cerro Rico, Potosí, Bolivia
- **Classification:** Declassified Q4 2025 (Report PAMC-PR-2025-011)
- **Outcome:** Successful
- **PAMC casualties:** 1 minor (Lt. Bauer-Cortez, hand burn)
- **Adversary casualties:** Cabrestante captured; 3 Iron Tributary affiliates captured
- **Notes:** Coordinated cell engagement utilizing segmented suspension (suspending the hydraulic arm at the elbow joint rather than the full augment). Demonstrated tactical workaround for above-ceiling targets. Cabrestante is currently in CBMP medical custody for augment study.

9.9.5 Case 2021-0091: Atacama Altiplano (Drift Read Confirmation)

- **Case number:** 2021-0091
- **Date:** 14–22 September 2021
- **Location:** Atacama altiplano, near Socaire, Antofagasta Region, Chile
- **Classification:** Declassified Q2 2022 (partial)
- **Outcome:** Partial
- **PAMC casualties:** none
- **Adversary casualties:** 1 Concession affiliate captured, 2 escaped
- **Notes:** First operationally validated deployment of the newly emerged Drift Read ability. The asset detected and interdicted three incoming small-arms rounds across a 36-hour engagement window. Drift Read was formally added to the asset's capability registry on 30 September 2021 following this case. CBMP collected the bulk of its early Drift Read telemetry during this operation.

9.9.6 Case 2024-0156: Tallinn-Asunción Liaison

- **Case number:** 2024-0156
- **Date:** 4–17 August 2024
- **Location:** Asunción, Paraguay (observer territory) and Tallinn, Estonia (joint EUMR coordination)
- **Classification:** Classified (Tier-2)
- **Outcome:** Successful
- **PAMC casualties:** none
- **Adversary casualties:** 3 captured (mixed Iron Tributary / EUMR adversary network); 1 fatality (non-PAMC operative)
- **Notes:** Rare extra-territorial deployment authorized by Director-General Espósito-Tagle. Joint operation with EUMR operatives **HOLLOWMARK** and **VERIDIAN**. The asset spent 9 days in Tallinn coordinating intelligence handoff. First instance of the asset's Inertial Suspension being demonstrated to non-PAMC liaisons under controlled observation. Produced a 14-page joint after-action report.

9.9.7 Case 2025-0220: Cordillera Cache Audit

- **Case number:** 2025-0220
- **Date:** 1–11 December 2025
- **Location:** All 19 cache locations across the asset's operational territory
- **Classification:** Routine (administrative)
- **Outcome:** Successful
- **PAMC casualties:** none
- **Adversary casualties:** none
- **Notes:** Triennial audit of all 19 micro-caches (§8.4). 3 caches required content rotation; 1 (Salar de Uyuni Sector boundary) was relocated 4.2 km due to restricted-zone expansion. No tampering evidence found. Audit completed under-budget and 2 days ahead of schedule.

9.10 Case 2026-0008: Helsinki Open — Bolivian Altiplano

- **Case number:** 2026-0008
- **Date:** 1 March 2026 – present
- **Location:** Bolivian altiplano (specifics classified)
- **Classification:** Open
- **Outcome:** ongoing
- **PAMC casualties:** none to date
- **Adversary:** suspected Mar Inverso (§7.10) and Iron Tributary mixed engagement
- **Notes:** Active investigation. Director-General authorization in place. Asset reports weekly to Director Salvatierra.

9.11 Summary Statistics (2019–2025)

Metric	Value
Total cases logged	47 (closed); 1 (open)
Successful outcomes	31 (66%)
Partial outcomes	6 (13%)
Inconclusive	5 (11%)
Pyrrhic	3 (6%)
Catastrophic	2 (4%)
PAMC operative KIA	1 (Cpl. Yáñez-Ríos, 2022)
Total adversary captures	92
Total civilian casualties (cumulative)	14 deaths, 34 injuries
Operational hours under power (career)	1,148
Combat-zone deployments	28
Non-combat / advisory deployments	19

Section 10. Glossary, Codenames, and Reference Tables

10.1 Consolidated Glossary

The following glossary consolidates terminology introduced in §§1–9. No new concepts are introduced.

- **Anchor Sling** — nylon-aramid harness used to redirect suspended objects in close quarters; 440 g; 1,200 kg static load rating. (§4.7)
- **Anchor Stone "Cerro-1"** — polished slate fragment recovered from the 2018 trigger site; 32 g; personal talisman. (§4.14)
- **Andean Operations Bureau** — PAMC bureau responsible for South American operations; compiling authority of this dossier. (§6.1)
- **Andina-7** — operative hardshell uniform, manufacturer Tejidos Cordillera, 3.8 kg. (§4.2)
- **Audiencia, The** — bureaucratic-adversary cluster within PAMC; estimated 7–12 members. (§7.11)
- **Bauer-Cortez, Solène** — Cell Halcón technical specialist; Class C-1 (EM sensing); ETH Zürich engineer. (§6.3)
- **Bersa-Nogales TX-9** — issued sidearm, 9×19 mm, 13-round magazine. (§4.5)
- **Bofedal-3** — PAMC-issued electrolyte solution; 740 mL/hr operational dose. (§2.14)
- **Borghi-Castelnuovo, Maite** — PAMC behavioral health, the asset's psychologist. (§6.4)
- **Cabrestante** — civilian Pablo Eyzaguirre-Pinto; cybernetically augmented adversary; Tier 3; captured 2025. (§7.7)
- **Calexin-7** — migraine prophylactic, 240 mg twice daily standard. (§2.15, §4.9)
- **Calvache-Marrón Equations** — theoretical framework underlying Inertial Suspension, published 11 May 2022. (§2.1)
- **Cardex-2** — restricted stimulant analog, 6 doses in medical kit. (§4.9)
- **CBMP** — Cooperative Bureau of Metahuman Physics, PAMC science directorate. (§2.1)
- **CEM** — Centro de Entrenamiento de Montaña, Río Blanco, Chilean Army mountain warfare school. (§1.3)
- **Cell Cóndor** — PAMC Mendoza-based cell; suffered casualty in 2022 Mendoza engagement. (§8.8, §9.4)
- **Cell Halcón** — SLATEFALL's 4-operative tactical unit; finalized 11 October 2022. (§6.2)
- **Cerro Castillo** — Aysén Region mountain, site of the 2018 trigger event; exclusion zone since 2020. (§3.4, §8.6)
- **Concession, The** — corporate-paramilitary syndicate, 62 confirmed members. (§7.9)
- **Cordillera Mk-IV** — issued combat-mountain boot, 980 g per boot. (§4.3)
- **Cooperative Service Agreement** — PAMC employment contract signed 12 March 2019. (§3.6)
- **Cuartel Valparaíso** — primary base, decommissioned naval depot, 4,820 m². (§8.1)
- **Doctrine of Sequential Suspension (DSS)** — formal combat doctrine, articulated 19 June 2022. (§5.1)
- **Drift Read** — secondary precognitive ability, 3-meter range, awakened May 2021. (§2.10)
- **Drop-Tune** — tactical manoeuvre exploiting the 0.3-second contact-exclusion clearance. (§5.5)
- **Echo Lock** — neurological failure mode, 4-second freeze; observed twice. (§2.13)
- **El Sereno** — unidentified nocturnal adversary, Tier 3, at-large. (§7.4)
- **EM-Suppression Cloak** — 1.2 kg aluminized weave, 24–28 dB attenuation. (§4.14)
- **Espósito-Tagle, María-Elena** — PAMC Director-General. (§6.1)
- **Faro-7** — issued headlamp, 720 lumens. (§4.10)
- **Four-Minute Window** — extraction-discipline standard; 4-minute time-to-extraction. (§5.10)
- **HALC-014** — the asset's customized loadout profile. (§4.1)
- **Helsinki Open** — informal name for Case 2026-0008. (§9.10)

- **Hold** — tertiary low-mass extended-duration ability; 2 kg max, 4 s max duration. (§2.11)
- **Inertial Suspension (IS)** — the asset's primary power. (§2.1)
- **Iron Tributary** — paramilitary cell, 22 confirmed members. (§7.8)
- **LMAF** — Lima Metahuman Assessment Facility. (§3.5)
- **Mar Inverso** — projected Tier 5 inertial-inversion threat. (§7.10)
- **Marrón-Saavedra, León** — CBMP physicist, co-author of the Calvache–Marrón Equations. (§6.4)
- **MARTILLO** — PAMC Class A-1 operative; documented professional friction with the asset. (§6.7)
- **Mendieta, Asunta** — safehouse keeper, Cuzco. (§6.5)
- **MIRADOR** — senior PAMC operative, the asset's former trainer. (§3.6, §8.8)
- **NMOC** — Nordic Metahuman Operations Cell; source of the Threat Tier Schedule format. (§7.1)
- **OEA-MAR** — OAS Misión Andina de Rescate, the asset's pre-PAMC deployment framework. (§1.4)
- **Operación Vilcabamba** — 2014 OEA-MAR Peruvian deployment. (§1.4)
- **Operación Trapecio** — 2016 OEA-MAR Bolivian deployment. (§1.4)
- **Operación Cancha Carrera** — 2017 OEA-MAR Argentine deployment. (§1.4)
- **Ovalle-Garrido, Ezequiel** — Chilean Army liaison officer, relationship since 2014. (§6.5)
- **PAMC** — Pan-American Metahuman Cooperative, parent organization. (§6.1)
- **PAMC Protocol 4-Bravo** — public-facing media restriction protocol. (§1.8)
- **PAMC Standing Order 9** — civilian-collateral zero-tolerance directive. (§5.8)
- **PAMC Restraint Facility, Iquique** — current custody site of Tropopausa. (§7.3)
- **Pajarera** — civilian intel asset Tessa Roehl, La Paz. (§6.5)
- **Quebrantadero** — civilian Wilma Sandoval-Iqui; bone-density manipulator; Tier 4; at-large. (§7.5)
- **Quiltro V-12** — armored utility vehicle, 4.4 t. (§4.11)
- **Quispe-Linares, Mateo** — Cell Halcón tracker, ex-Bolivian SF. (§6.3)
- **Renström-Verdugo, Catalina** — the asset's spouse, pediatric oncologist. (§1.7)
- **Reset Bracers** — neural-telemetry wrist units, 180 g each. (§4.6)
- **Salvatierra-Bunge, Aída** — PAMC Operations Director. (§6.4)
- **Sletten, Magnus** — PAMC Inspector General, "Underwriter." (§7.6)
- **SLATEFALL** — registered codename of the asset, assigned 7 May 2019. (§1.1)
- **Suspended Field** — tactical multi-baton kinetic-cover technique. (§5.6)
- **Tail Momentum** — the property whereby suspended objects resume exact prior vector on release. (§2.6)
- **Tail-Strike** — combat technique exploiting Tail Momentum. (§5.4)
- **Three-Two-One Rule** — tactical heuristic (3 activations / 2 sightlines / 1 extraction). (§5.2)
- **Tropopausa** — civilian Eliseo Mardones-Vodanovic; pressure manipulator; Tier 4; contained. (§7.3)
- **Underwriter** — bureaucratic adversary, Magnus Sletten. (§7.6)
- **Vendrame, Raúl** — Cell Halcón medic. (§6.3)
- **Yatiri-3** — burst-radio handset, 460 g. (§4.8)
- **0-3-Castillo** — asset's radio call sign. (§1.1)

10.2 Codename Index

Codename	True identity	Affiliation
SLATEFALL	Inez Calvache-Renström	PAMC (asset)
MIRADOR	Estefania Pizarroso-Aldama	PAMC (allied)
MARTILLO	Joaquín Hidalgo-Yáñez	PAMC (rival)

Codename	True identity	Affiliation
Pajarera	Tessa Roehl	Civilian intel asset
Tropopausa	Eliseo Mardones-Vodanovic	Adversary
El Sereno	Unknown	Adversary
Quebrantadero	Wilma Sandoval-Iqui	Adversary
Cabrestante	Pablo Eyzaguirre-Pinto	Adversary (captured)
Underwriter	Magnus Sletten	Adversary (bureaucratic)
Cataract	Helene Drovac	Adversary (Concession contract)
Mar Inverso	Unknown	Adversary (projected)

10.3 Threat Tier Reference Table (Repeated for Convenience)

Tier	Power output ceiling	Casualty potential	Authorization required
1	< 0.5 kJ	≤ 1 civilian	Field operative, solo
2	0.5–25 kJ	2–10 civilians	Field officer, solo or cell
3	25–500 kJ	11–100 civilians	Lead officer, cell required
4	500 kJ – 50 MJ	101–5,000 civilians	Lead officer + auxiliary
5	> 50 MJ or strategic	> 5,000 civilians	Director-General authorization

10.4 Power-Mechanics Reference Card

Parameter	Value
Targeting time	1.2 s
Mass ceiling	240 kg
Uncontrolled peak	380 kg (2018 only)
Baseline duration	0.6 s
Pharmacologically extended duration	1.1 s (Calexin-7, 240 mg)
Effective range	22 m
Theoretical range	41 m
Per-target cooldown	11 s
Global activation cooldown	2.4 s
Activations per minute (sustained)	3
Effective inertial reduction	~99.6% (to 0.4% rest mass)

Parameter	Value
Drift Read range	3.0 m
Drift Read lead-time	~0.30 s
Hold mass max	2 kg
Hold duration max	4.0 s
Acoustic-disruption threshold	≥ 135 dB
Echo Lock trigger threshold	≥ 145 dB / > 240 kg / glucose < 58 mg/dL
Operational metabolic cost	2,100 kcal/hr

10.5 Equipment Quick-Reference

Item	Mass	Manufacturer
Andina-7 hardshell	3.8 kg	Tejidos Cordillera
Cordillera Mk-IV boot (single)	980 g	Calzados Pirque
Tungsten throwing baton (single)	340 g	Talleres Magallanes
Bersa-Nogales TX-9	720 g loaded	Bersa-Nogales
Reset Bracer (single)	180 g	CBMP / Bauer-Cortez
Anchor Sling	440 g	Tejidos Cordillera
Yatiri-3 handset	460 g	PAMC Signals
Quiltro V-12 vehicle	4,400 kg	Industrias Ñuble
EM-Suppression Cloak	1.2 kg	CBMP

10.6 Timeline Master Table (2018–2026)

Year	Key events
2018	Cerro Castillo trigger (8 Oct)
2019	PAMC service agreement (12 Mar); codename assigned (7 May)
2020	First Echo Lock event (19 Nov)
2021	Drift Read awakens, Antofagasta (17 May)
2022	Calvache–Marrón Equations published (11 May); Mendoza Pyrrhic (3 Jun); DSS articulated (19 Jun); promoted F.O. Grade 2 (1 Aug); Cell Halcón finalized (11 Oct)
2023	Second Echo Lock, Salta (14 Aug); marriage (4 Nov)
2024	Promoted Lead Officer Grade 3 (9 Feb); Tarapacá raid (17 Feb)

Year	Key events
2025	Audiencia counter-audit (Jan–Feb); Cabrestante neutralization (18 Sep); Mar Inverso intel (22 Dec)
2026	Helsinki Open ongoing (1 Mar–present); current revision date (11 Apr)

10.7 Adversary–Incident Cross-Reference Matrix

Adversary	Cases involved
Tropopausa	2022-0123, 2023 (Antofagasta), 2023 (Iquique), 2024 (Coquimbo)
El Sereno	2023-0067, 2024 (Antofagasta), 2025 (Arica)
Quebrantadero	2024-0202, 2024 (Cuzco)
Cabrestante	2025-0156
Underwriter	2025-0019
Iron Tributary	2024-0034, 2025-0156
The Concession	2024-0034 (primary), 4 prior contact engagements
Mar Inverso	2026-0008 (suspected)

10.8 Mark-Degradation and Power-Failure Reference

Failure factor	Threshold	Effect
Acoustic shock	135 dB	Targeting interruption
Acoustic shock	145 dB	Echo Lock trigger
Hypothermia	core < 34.6 °C	Power disabled
Hypoglycemia	glucose < 58 mg/dL	Echo Lock risk
Hyperventilation	> 34 breaths/min, > 90 s	18% targeting degradation
EM-pulse	> 0.4 T	Secondary nerve dysfunction
Visual occlusion	> 40 min sustained	Transient suppression
Altitude	> 5,800 m	(Tropopausa-specific; not the asset's own limit)

10.9 Acronym Key

Acronym	Expansion
PAMC	Pan-American Metahuman Cooperative
CBMP	Cooperative Bureau of Metahuman Physics
CEM	Centro de Entrenamiento de Montaña

Acronym	Expansion
ESM	Escuela Militar Bernardo O'Higgins
LMAF	Lima Metahuman Assessment Facility
NMOC	Nordic Metahuman Operations Cell (Threat Tier source)
OEA-MAR	Organización de los Estados Americanos — Misión Andina de Rescate
DSS	Doctrine of Sequential Suspension
IS	Inertial Suspension
DAM	Dirección de Asuntos Metahumanos (Peruvian agency)
SOE	Servicio de Operaciones Especiales (Bolivian agency)
MTBF	Mean Time Between Failures
IIIA / IV	NIJ ballistic protection ratings

10.10 Document Provenance and Revision History

Revision	Date	Editor	Notes
1.0	7 May 2019	Andean Operations Bureau	Initial filing post-codename assignment
2.0	14 Jan 2020	M. Rosenblatt-Velarde	Post-probation update
3.0	22 Aug 2020	M. Rosenblatt-Velarde	Echo Lock event added
4.0	30 Jun 2021	A. Quintero-Salgado	Antofagasta + Drift Read addition
5.0	19 Jun 2022	A. Quintero-Salgado	DSS articulation added
6.0	31 Aug 2022	A. Quintero-Salgado	Promotion and Cell Halcón
7.0	12 Mar 2023	A. Quintero-Salgado	Cell composition finalized
8.0	24 Sep 2023	M.-J. Casariego-Núñez	Salta inquiry findings
9.0	15 Feb 2024	M.-J. Casariego-Núñez	Lead-officer promotion
10.0	28 Apr 2024	M.-J. Casariego-Núñez	Tarapacá declassification
11.0	30 Sep 2024	A. Quintero-Salgado	Annual review cycle
12.0	28 Feb 2025	A. Quintero-Salgado	Audiencia outcome added
13.0	30 Sep 2025	A. Quintero-Salgado	Cabrestante engagement
14.0	28 Dec 2025	A. Quintero-Salgado	Mar Inverso intel added
14.1	28 Dec 2025	A. Quintero-Salgado	Minor corrections
14.2	11 Apr 2026	A. Quintero-Salgado	Current revision; Helsinki Open update

10.11 Document Closure

This dossier constitutes the official PAMC reference file for asset **PAMC-A2-014 (SLATEFALL)** as of 11 April 2026. All factual claims herein are derived from CBMP scientific records, PAMC Operations Directorate after-action filings, and primary debrief sessions with the asset. Discrepancies between this revision and prior revisions should be resolved in favor of the most recent revision unless explicitly noted.

Inquiries regarding the contents of this dossier should be directed to the **Andean Operations Bureau, PAMC South Cone Headquarters, Cuartel Valparaíso**.

— *End of Dossier* —